

Homework 1

This project will be evaluating the following double inverted pendulum system:

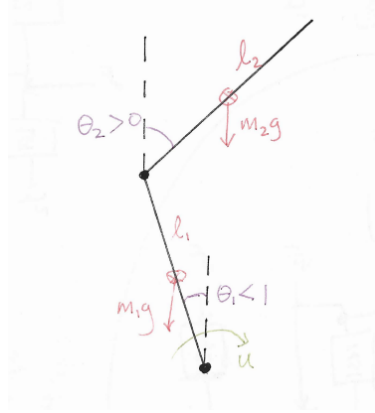


Figure 1: Double inverted pendulum.

An input torque u is applied to the first link of the pendulum. Each arm is modeled as a uniform rod and friction between the rods is assumed as $|F_1| = b_1|\dot{\theta}_1|$ and $|F_2| = b_2|\dot{\theta}_1 - \dot{\theta}_2|$. Using the angular deviation from the vertical and angular velocities of the links, we can define the state $x = [\theta_1 \ \theta_2 \ \dot{\theta}_1 \ \dot{\theta}_2]^T$. Using Lagrangian mechanics, the dynamics of the system can be expressed using the following first order differential equation:

$$\begin{aligned}
 & \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & (\frac{1}{3}m_1 + m_2)l_1^2 & \frac{1}{2}m_2l_1l_2\cos(\theta_1 - \theta_2) \\ 0 & 0 & \frac{1}{2}m_2l_1l_2\cos(\theta_1 - \theta_2) & \frac{1}{3}m_2l_2^2 \end{bmatrix} \begin{bmatrix} \dot{\theta}_1 \\ \dot{\theta}_2 \\ \ddot{\theta}_1 \\ \ddot{\theta}_2 \end{bmatrix} \\
 &= \begin{bmatrix} 0 \\ 0 \\ -\frac{1}{2}m_2l_1l_2\sin(\theta_1 - \theta_2)\dot{\theta}_2^2 + (\frac{1}{2}m_1 + m_2)gl_1\sin\theta_1 \\ \frac{1}{2}m_2l_1l_2\sin(\theta_1 - \theta_2)\dot{\theta}_1^2 + \frac{1}{2}m_2gl_2\sin\theta_2 \end{bmatrix} + \begin{bmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & -(b_1 + b_2) & b_2 \\ 0 & 0 & b_2 & -b_2 \end{bmatrix} \begin{bmatrix} \theta_1 \\ \theta_2 \\ \dot{\theta}_1 \\ \dot{\theta}_2 \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ u \\ 0 \end{bmatrix}
 \end{aligned}$$

1 Existence and uniqueness of solutions

1.1 Local Lipschitz continuity

For the evaluation of the solutions to this system, we will let the inputs of the system be zero. To verify the existence and uniqueness of solutions, it would be useful to apply Theorem 2.3. Proving local Lipschitz continuity would demonstrate that unique solutions to our IVP exist. If we can prove that our first order ODE is continuously differentiable, we can use Theorem 2.2 to demonstrate local Lipschitz continuity and, thus, the existence of solutions.

Let us first rewrite our dynamics equation into the form $\dot{x} = f(x)$.

$$M(x) = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & (\frac{1}{3}m_1 + m_2)l_1^2 & \frac{1}{2}m_2l_1l_2 \cos(\theta_1 - \theta_2) \\ 0 & 0 & \frac{1}{2}m_2l_1l_2 \cos(\theta_1 - \theta_2) & \frac{1}{3}m_2l_2^2 \end{bmatrix}$$

$$\det M(x) = \frac{1}{9}m_1m_2l_1^2l_2^2 + \frac{1}{3}m_2^2l_1^2l_2^2 \sin^2(\theta_1 - \theta_2) + \frac{1}{12}m_2^2l_1^2l_2^2 \cos^2(\theta_1 - \theta_2) > 0$$

$M(x)$ is invertible for all $x \in \mathbb{R}^4$, thus we can compute its inverse.

$$M^{-1}(x) = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & \frac{\frac{1}{3}m_2l_2^2}{|M(x)|} & -\frac{\frac{1}{2}m_2l_1l_2 \cos(\theta_1 - \theta_2)}{|M(x)|} \\ 0 & 0 & -\frac{\frac{1}{2}m_2l_1l_2 \cos(\theta_1 - \theta_2)}{|M(x)|} & \frac{(\frac{1}{3}m_1 + m_2)l_1^2}{|M(x)|} \end{bmatrix}$$

$$= \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & \frac{\mu_1}{|M(x)|} & \frac{\mu_3(x)}{|M(x)|} \\ 0 & 0 & \frac{\mu_3(x)}{|M(x)|} & \frac{\mu_2}{|M(x)|} \end{bmatrix}$$

Let us also rewrite the right-hand side of our differential equation.

$$\begin{bmatrix} 0 \\ 0 \\ -\frac{1}{2}m_2l_1l_2 \sin(\theta_1 - \theta_2) \dot{\theta}_2^2 + (\frac{1}{2}m_1 + m_2)gl_1 \sin \theta_1 \\ \frac{1}{2}m_2l_1l_2 \sin(\theta_1 - \theta_2) \dot{\theta}_1^2 + \frac{1}{2}m_2gl_2 \sin \theta_2 \end{bmatrix} + \begin{bmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & -(b_1 + b_2) & b_2 \\ 0 & 0 & b_2 & -b_2 \end{bmatrix} \begin{bmatrix} \theta_1 \\ \theta_2 \\ \dot{\theta}_1 \\ \dot{\theta}_2 \end{bmatrix}$$

$$= \begin{bmatrix} \dot{\theta}_1 \\ \dot{\theta}_2 \\ -\frac{1}{2}m_2l_1l_2 \sin(\theta_1 - \theta_2) \dot{\theta}_2^2 + (\frac{1}{2}m_1 + m_2)gl_1 \sin \theta_1 - (b_1 + b_2)\dot{\theta}_1 + b_2\dot{\theta}_2 \\ \frac{1}{2}m_2l_1l_2 \sin(\theta_1 - \theta_2) \dot{\theta}_1^2 + \frac{1}{2}m_2gl_2 \sin \theta_2 + b_2\dot{\theta}_1 - b_2\dot{\theta}_2 \end{bmatrix} = \begin{bmatrix} \nu_1(x) \\ \nu_2(x) \\ \nu_3(x) \\ \nu_4(x) \end{bmatrix}$$

Our state space equation can now be rewritten:

$$\dot{x} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & \frac{\mu_1}{|M(x)|} & \frac{\mu_3(x)}{|M(x)|} \\ 0 & 0 & \frac{\mu_3(x)}{|M(x)|} & \frac{\mu_2}{|M(x)|} \end{bmatrix} \begin{bmatrix} \nu_1(x) \\ \nu_2(x) \\ \nu_3(x) \\ \nu_4(x) \end{bmatrix}$$

$$= \begin{bmatrix} \nu_1(x) \\ \nu_2(x) \\ \frac{\mu_1\nu_3(x) + \mu_3(x)\nu_4(x)}{|M(x)|} \\ \frac{\mu_3(x)\nu_3(x) + \mu_2\nu_4(x)}{|M(x)|} \end{bmatrix} = \begin{bmatrix} f_1(x) \\ f_2(x) \\ \frac{f_3(x)}{|M(x)|} \\ \frac{f_4(x)}{|M(x)|} \end{bmatrix} = f(x)$$

Computing the Jacobian $Df(x)$:

$$Df(x) = \begin{bmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ \frac{\partial}{\partial \theta_1} \frac{f_3(x)}{|M(x)|} & \frac{\partial}{\partial \theta_2} \frac{f_3(x)}{|M(x)|} & \frac{\partial}{\partial \theta_1} \frac{f_3(x)}{|M(x)|} & \frac{\partial}{\partial \theta_2} \frac{f_3(x)}{|M(x)|} \\ \frac{\partial}{\partial \theta_1} \frac{f_4(x)}{|M(x)|} & \frac{\partial}{\partial \theta_2} \frac{f_4(x)}{|M(x)|} & \frac{\partial}{\partial \theta_1} \frac{f_4(x)}{|M(x)|} & \frac{\partial}{\partial \theta_2} \frac{f_4(x)}{|M(x)|} \end{bmatrix}$$

Each element in the lower half of the Jacobian can be computed using the quotient rule. The numerators are sums of products of sinusoids and polynomial functions. The denominators equal $|M(x)|^2$:

$$|M(x)|^2 = \left\{ \frac{1}{3} \left(\frac{1}{3} m_1 + m_2 \right) m_2 l_1^2 l_2^2 - \frac{1}{4} m_2^2 l_1^2 l_2^2 \cos^2(\theta_1 - \theta_2) \right\}^2$$

$|M(x)|^2 \neq 0$ so long as $m_1, m_2, l_1, l_2 \neq 0$, which are reasonable restrictions for the physical and mechanical nature of this problem. Because the denominator is composed of constant functions and sinusoids and can never be zero, $\frac{\partial f_n}{\partial x_m}$ will always exist and will be a continuous function across all $x \in E$.

Therefore, $f(x)$ is a continuously differentiable function, we can apply Theorem 2.2 and conclude that $f(x)$ is also **locally Lipschitz continuous**. Furthermore, we can thus apply Theorem 2.3 and declare that there exists $a \in \mathbb{R}_{>0}$ such that there exists a unique solution to the IVP for our system within an interval $I = (t_0 - a, t_0 + a)$.

1.2 Global Lipschitz continuity

Global Lipschitz continuity of $f(x)$ depends on its domain, E . In theory, $f(x)$ operates on any input $x \in \mathbb{R}^4$. In this case, $f(x)$ is not globally Lipschitz continuous, as there does not exist an $L \in \mathbb{R}_{>0}$ for which the inequality

$$\|f(x) - f(y)\| \leq L\|x - y\|$$

holds true. This is true because there are quadratic terms of the state (i.e. $\dot{\theta}_1^2$ and $\dot{\theta}_2^2$) present in $f(x)$. The derivatives of these functions are not bounded as these states become infinitely large. As a result, there is no finite Lipschitz constant that can make the above inequality hold true.

However, if we limit the domain to physically reasonable (i.e. finite angular velocities) then $f(x)$ does become Lipschitz continuous over this bounded domain. In this case, the solutions to the IVP are strengthened by Corollary 2.1, in which a unique solution exists on an interval $I = (t_0 - a, t_0 + a)$ for any $a \in \mathbb{R}_{>0}$.

2 Continuity of solutions

We can consider the case where we have bounded domain for $f(x)$. Lipschitz continuity on $f(x)$ means that Theorem 3.2 applies to our system. Since we expect that it is continuous with respect to initial conditions, it would obey the inequality

$$\|x(t) - z(t)\| \leq \|x_0 - z_0\| e^{L(t-t_0)}, \forall t \in [t_0, t_1]$$

We can attempt to verify this numerically. The first step is to identify the Lipschitz constant L for the bounding equation. We can attempt to define L as

$$L = \max_{x \in E} \|Df(x)\|$$

However, because $\|Df(x)\|$ becomes arbitrarily large as the angular velocities increase, the above inequality will generally be true for any initial conditions near x_0 —

a sufficiently sized L can be chosen. This is cool to know, but isn't quite helpful if we want to do numerical verifications.

Something we can try to do is restrict the set of x for which we evaluate $\|Df(x)\|$. I have restricted x to be the set of states that the system passes through from an initial condition x_0 . This seems like a valid approach to me — since we expect the solutions to be close around a specific solution, the continuous nature of $Df(x)$ lets us assume these solutions have similar Jacobians.

Implementing this into MATLAB, we can use the `ode45` function to simulate our IVP. Furthermore we can use the `jacobian` function from the Symbolic Toolbox to calculate $\|Df(x)\|_2$ along the path of the initial solution and select its maximum value as L . We can pick another initial condition z_0 around x_0 and simulate its IVP. We can then check the error against the right hand side of the inequality.

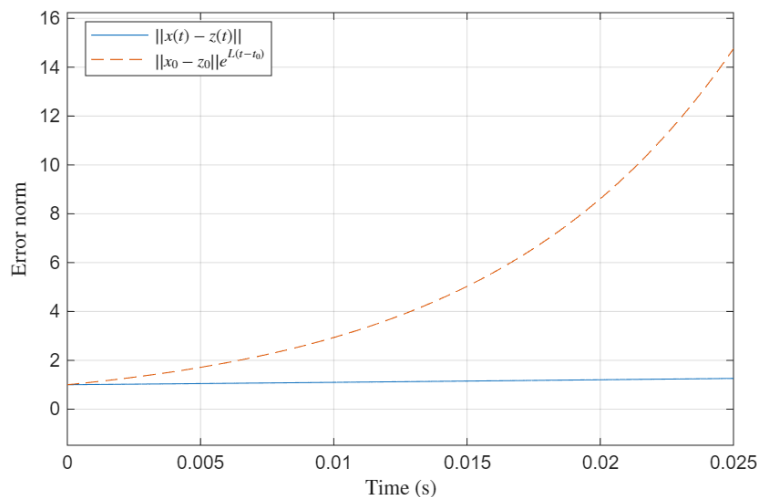


Figure 2: Error norm of two solutions to two IVPs compared to theoretical upper bound.

We can repeat this a number of times for the same value of $\|x_0 - z_0\|$ and see if the inequality holds.

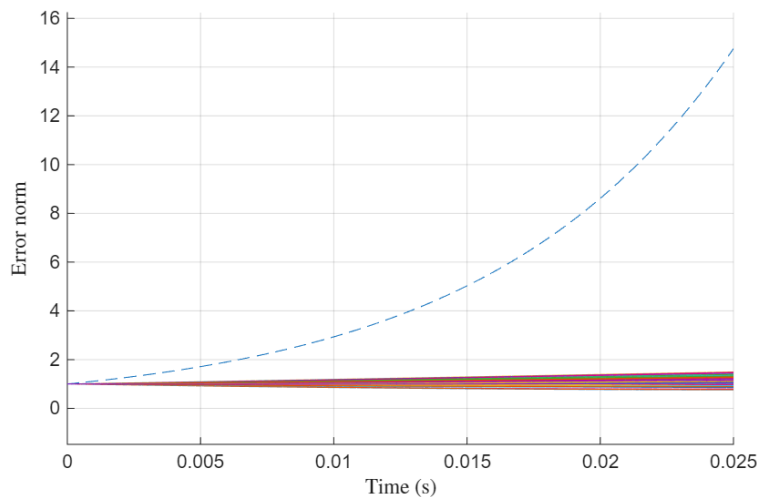


Figure 3: Error norm of solutions to 500 IVPs. $\|x_0 - z_0\| = 1$ for all trials.

We can see that the solutions are well below the theoretical upper bound set by the inequality. In its full form, Theorem 3.2 has a bounding inequality of

$$\|x(t) - z(t)\| \leq \|x_0 - z_0\|e^{L(t-t_0)} + \mu(t-t_0)e^{L(t-t_0)}, \forall t \in [t_0, t_1]$$

$\mu(t-t_0)e^{L(t-t_0)} \in \mathbb{R}_{\geq 0}$, so this inequality also holds for our system.

Homework 2

This project will be evaluating the following double inverted pendulum system:

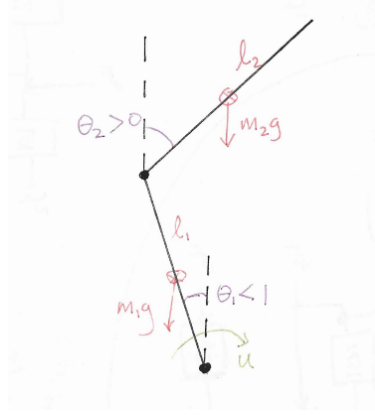


Figure 1: Double inverted pendulum.

An input torque u is applied to the first link of the pendulum. Each arm is modeled as a uniform rod ($I_i = \frac{1}{12}m_i l_i^2$) and friction between the rods is assumed as $|F_1| = b_1|\dot{\theta}_1|$ and $|F_2| = b_2|\dot{\theta}_1 - \dot{\theta}_2|$. Using the angular deviation from the vertical and angular velocities of the links, we can define the state $x = [\theta_1 \ \theta_2 \ \dot{\theta}_1 \ \dot{\theta}_2]^\top$. The dynamics of the system are driven by the following differential equation in the form $\dot{x} = f(x, u)$:

$$\dot{x} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & \mu_2(x) & \mu_3(x) \\ 0 & 0 & \mu_3(x) & \mu_1(x) \end{bmatrix} \left(\begin{bmatrix} \nu_1(x) \\ \nu_2(x) \\ \nu_3(x) \\ \nu_4(x) \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ 1 \\ 0 \end{bmatrix} u \right) = \begin{bmatrix} \nu_1(x) \\ \nu_2(x) \\ \mu_2(x)\nu_3(x) + \mu_3(x)\nu_4(x) \\ \mu_3(x)\nu_3(x) + \mu_1(x)\nu_4(x) \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ \mu_2(x) \\ \mu_3(x) \end{bmatrix} u$$

where:

$$\begin{aligned} \mu_2(x) &= \frac{M_2}{|M(x)|}, \quad \mu_3(x) = \frac{-M_3 \cos \Delta}{|M(x)|}, \quad \mu_1(x) = \frac{M_1}{|M(x)|} \\ \nu_1(x) &= \dot{\theta}_1, \quad \nu_2(x) = \dot{\theta}_2 \\ \nu_3(x) &= -K_1 \dot{\theta}_2^2 \sin \Delta + K_2 \sin \theta_1 - B_1 \dot{\theta}_1 + B_2 \dot{\theta}_2 \\ \nu_4(x) &= K_3 \dot{\theta}_1^2 \sin \Delta + K_4 \sin \theta_2 + B_2 \dot{\theta}_1 - B_2 \dot{\theta}_2 \end{aligned}$$

with parameters defined in Table 1.

M_1	$(\frac{1}{3}m_1 + m_2) l_1^2$
M_2	$\frac{1}{3}m_2 l_2^2$
M_3	$\frac{1}{2}m_2 l_1 l_2$
K_1	$\frac{1}{2}m_2 l_1 l_2$
K_2	$(\frac{1}{2}m_1 + m_2) g l_1$
K_3	$\frac{1}{2}m_2 l_1 l_2$
K_4	$\frac{1}{2}m_2 g l_2$
B_1	$(b_1 + b_2)$
B_2	b_2
Δ	$\theta_1 - \theta_2$
$ M(x) $	$M_1 M_2 - M_3^2 \cos^2 \Delta$

Table 1: Coefficients for rest of derivations.

The Jacobian of the system is:

$$Df(x) = \begin{bmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ Df_{31}(x) & Df_{32}(x) & Df_{33}(x) & Df_{34}(x) \\ Df_{41}(x) & Df_{42}(x) & Df_{43}(x) & Df_{44}(x) \end{bmatrix}$$

where the elements are defined by the equations below:

$$\begin{aligned} Df_{31}(x) &= \frac{\partial \mu_2(x)}{\partial \theta_1} \nu_3(x) + \mu_2(x) \frac{\partial \nu_3(x)}{\partial \theta_1} + \frac{\partial \mu_3(x)}{\partial \theta_1} \nu_4(x) + \mu_3(x) \frac{\partial \nu_4(x)}{\partial \theta_1} \\ Df_{32}(x) &= \frac{\partial \mu_2(x)}{\partial \theta_2} \nu_3(x) + \mu_2(x) \frac{\partial \nu_3(x)}{\partial \theta_2} + \frac{\partial \mu_3(x)}{\partial \theta_2} \nu_4(x) + \mu_3(x) \frac{\partial \nu_4(x)}{\partial \theta_2} \\ Df_{33}(x) &= \mu_2(x) \frac{\partial \nu_3(x)}{\partial \dot{\theta}_1} + \mu_3(x) \frac{\partial \nu_4(x)}{\partial \dot{\theta}_1} \\ Df_{34}(x) &= \mu_2(x) \frac{\partial \nu_3(x)}{\partial \dot{\theta}_2} + \mu_3(x) \frac{\partial \nu_4(x)}{\partial \dot{\theta}_2} \\ Df_{41}(x) &= \frac{\partial \mu_3(x)}{\partial \theta_1} \nu_3(x) + \mu_3(x) \frac{\partial \nu_3(x)}{\partial \theta_1} + \frac{\partial \mu_1(x)}{\partial \theta_1} \nu_4(x) + \mu_1(x) \frac{\partial \nu_4(x)}{\partial \theta_1} \\ Df_{42}(x) &= \frac{\partial \mu_3(x)}{\partial \theta_2} \nu_3(x) + \mu_3(x) \frac{\partial \nu_3(x)}{\partial \theta_2} + \frac{\partial \mu_1(x)}{\partial \theta_2} \nu_4(x) + \mu_1(x) \frac{\partial \nu_4(x)}{\partial \theta_2} \\ Df_{43}(x) &= \mu_3(x) \frac{\partial \nu_3(x)}{\partial \dot{\theta}_1} + \mu_1(x) \frac{\partial \nu_4(x)}{\partial \dot{\theta}_1} \\ Df_{44}(x) &= \mu_3(x) \frac{\partial \nu_3(x)}{\partial \dot{\theta}_2} + \mu_1(x) \frac{\partial \nu_4(x)}{\partial \dot{\theta}_2} \end{aligned}$$

$$D \begin{bmatrix} \mu_1(x) \\ \mu_2(x) \\ \mu_3(x) \end{bmatrix} = \begin{bmatrix} -M_1 M_3 \frac{\cos \Delta \sin \Delta}{|M(x)|^2} & M_1 M_3 \frac{\cos \Delta \sin \Delta}{|M(x)|^2} & 0 & 0 \\ -M_2 M_3 \frac{\cos \Delta \sin \Delta}{|M(x)|^2} & M_2 M_3 \frac{\cos \Delta \sin \Delta}{|M(x)|^2} & 0 & 0 \\ -M_3 \frac{(2M_3 \cos^2 \Delta - |M(x)|) \sin \Delta}{|M(x)|^2} & M_3 \frac{(2M_3 \cos^2 \Delta - |M(x)|) \sin \Delta}{|M(x)|^2} & 0 & 0 \end{bmatrix}$$

$$D \begin{bmatrix} \nu_1(x) \\ \nu_2(x) \\ \nu_3(x) \\ \nu_4(x) \end{bmatrix} = \begin{bmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ -K_1 \dot{\theta}_2^2 \cos \Delta + K_2 \cos \theta_1 & K_1 \dot{\theta}_2^2 \cos \Delta & -B_1 & -2K_1 \dot{\theta}_2 \sin \Delta + B_2 \\ K_3 \dot{\theta}_1^2 \cos \Delta & -K_3 \dot{\theta}_1^2 \cos \Delta + K_4 \cos \theta_2 & 2K_3 \dot{\theta}_1 \sin \Delta + B_2 & -B_2 \end{bmatrix}$$

1 System equilibria

We can determine the set of equilibrium points $\{x_i^*\}$ of the system by setting inputs to zero and determining the states for which $f(x, 0) = 0$ (Definition 4.5).

$$f(x, 0) = \begin{bmatrix} \nu_1(x) \\ \nu_2(x) \\ \mu_2(x)\nu_3(x) + \mu_3(x)\nu_4(x) \\ \mu_3(x)\nu_3(x) + \mu_1(x)\nu_4(x) \end{bmatrix} = 0$$

$$\Rightarrow \nu_1(x) = \dot{\theta}_1 = 0$$

$$\Rightarrow \nu_2(x) = \dot{\theta}_2 = 0$$

$$\mu_2(x)\nu_3(x) + \mu_3(x)\nu_4(x) \Big|_{\dot{\theta}_1, \dot{\theta}_2=0} = \frac{M_2 K_2 \sin \theta_1 - M_3 K_4 \cos \Delta \sin \theta_2}{|M(x)|} = 0$$

$$\mu_3(x)\nu_3(x) + \mu_1(x)\nu_4(x) \Big|_{\dot{\theta}_1, \dot{\theta}_2=0} = \frac{-M_3 K_2 \cos \Delta \sin \theta_1 + M_1 K_4 \sin \theta_2}{|M(x)|} = 0$$

$|M(x)| > 0 \forall x \in E$, so we can focus on the numerators only for the remaining terms.

$$\begin{bmatrix} M_2 K_2 & -M_3 K_4 \cos \Delta \\ -M_3 K_2 \cos \Delta & M_1 K_4 \end{bmatrix} \begin{bmatrix} \sin \theta_1 \\ \sin \theta_2 \end{bmatrix} = 0$$

The above equation hold true when either: 1) $\begin{bmatrix} \sin \theta_1 & \sin \theta_2 \end{bmatrix}^\top = 0$ or 2) when the matrix on the left has a null space. In the first case, the solutions are $\theta_1 = n\pi$, $\theta_2 = m\pi$ where $n, m \in \mathbb{Z}$. In latter case, we check when the determinant of the matrix becomes zero.

$$\det \begin{bmatrix} M_2 K_2 & -M_3 K_4 \cos \Delta \\ -M_3 K_2 \cos \Delta & M_1 K_4 \end{bmatrix} = K_2 K_4 M_1 M_2 - K_2 K_4 M_3^2 \cos^2 \Delta = 0$$

$$M_1 M_2 - M_3^2 \cos^2 \Delta = 0$$

$$|M(x)| = 0$$

We have shown before that $|M(x)| > 0 \forall x \in E$, thus only the solutions to Case 1 are the equilibrium points of the system.

$$\{x_i^*\} = \left\{ x \in E \mid x = \begin{bmatrix} 0 \\ 0 \\ n\pi \\ m\pi \end{bmatrix} \mid n, m \in \mathbb{Z} \right\}$$

Though this set may have many solutions, because the angles wrap every 2π radians, each equilibrium can add or subtract 2π to its angular positions and remain equivalent

to an equilibrium where $\theta_1, \theta_2 \in [0, 2\pi)$. Therefore, we can focus our analysis on four equilibria:

$$x_1^* = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}, x_2^* = \begin{bmatrix} \pi \\ 0 \\ 0 \\ 0 \end{bmatrix}, x_3^* = \begin{bmatrix} 0 \\ \pi \\ 0 \\ 0 \end{bmatrix}, x_4^* = \begin{bmatrix} \pi \\ \pi \\ 0 \\ 0 \end{bmatrix}$$

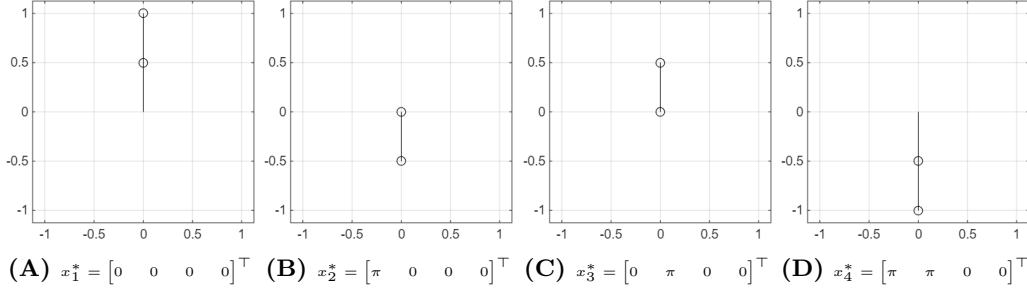


Figure 2: System equilibria of interest.

2 System linearization

2.1 General system

If we want to use linearization to analyze this system, we should linearize the system around each equilibrium point via $\dot{x} = Df(x_i^*)x = A_i x$.

When evaluating the Jacobian, many parameters are the same between equilibria, such as $|M(x^*)| = M_1 M_2 - M_3^2$. The values that *do* change have the values seen in Table 2.

	x_1^*	x_2^*	x_3^*	x_4^*
$\mu_3(x^*)$	$\frac{-M_3}{M_1 M_2 - M_3^2}$	$\frac{M_3}{M_1 M_2 - M_3^2}$	$\frac{M_3}{M_1 M_2 - M_3^2}$	$\frac{-M_3}{M_1 M_2 - M_3^2}$
$\frac{\partial \nu_3}{\partial \theta_1}(x^*)$	K_2	$-K_2$	K_2	$-K_2$
$\frac{\partial \nu_4}{\partial \theta_2}(x^*)$	K_4	K_4	$-K_4$	$-K_4$

Table 2: Relevant values of the Jacobian that change between equilibria.

$$A_1 = \begin{bmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ \frac{M_2 K_2}{|M(x^*)|} & \frac{-M_3 K_4}{|M(x^*)|} & \frac{-M_2 B_1 - M_3 B_2}{|M(x^*)|} & \frac{M_2 B_2 + M_3 B_2}{|M(x^*)|} \\ \frac{-M_3 K_2}{|M(x^*)|} & \frac{M_1 K_4}{|M(x^*)|} & \frac{M_3 B_1 + M_1 B_2}{|M(x^*)|} & \frac{-M_3 B_2 - M_1 B_2}{|M(x^*)|} \end{bmatrix}$$

$$A_2 = \begin{bmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ \frac{-M_2 K_2}{|M(x^*)|} & \frac{M_3 K_4}{|M(x^*)|} & \frac{-M_2 B_1 + M_3 B_2}{|M(x^*)|} & \frac{M_2 B_2 - M_3 B_2}{|M(x^*)|} \\ \frac{-M_3 K_2}{|M(x^*)|} & \frac{M_1 K_4}{|M(x^*)|} & \frac{-M_3 B_1 + M_1 B_2}{|M(x^*)|} & \frac{M_3 B_2 - M_1 B_2}{|M(x^*)|} \end{bmatrix}$$

$$A_3 = \begin{bmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ \frac{M_2 K_2}{|M(x^*)|} & \frac{-M_3 K_4}{|M(x^*)|} & \frac{-M_2 B_1 + M_3 B_2}{|M(x^*)|} & \frac{M_2 B_2 - M_3 B_2}{|M(x^*)|} \\ \frac{M_3 K_2}{|M(x^*)|} & \frac{-M_1 K_4}{|M(x^*)|} & \frac{-M_3 B_1 + M_1 B_2}{|M(x^*)|} & \frac{M_3 B_2 - M_1 B_2}{|M(x^*)|} \end{bmatrix}$$

$$A_4 = \begin{bmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ \frac{-M_2 K_2}{|M(x^*)|} & \frac{M_3 K_4}{|M(x^*)|} & \frac{-M_2 B_1 - M_3 B_2}{|M(x^*)|} & \frac{M_2 B_2 + M_3 B_2}{|M(x^*)|} \\ \frac{M_3 K_2}{|M(x^*)|} & \frac{-M_1 K_4}{|M(x^*)|} & \frac{M_3 B_1 + M_1 B_2}{|M(x^*)|} & \frac{-M_3 B_2 - M_1 B_2}{|M(x^*)|} \end{bmatrix}$$

If we want to get a sense of stability of these systems via Fact 6.1, we need to establish that: 1) these equilibria are isolated and 2) the real components of their eigenvalues are strictly less than zero.

The first requirement can be easily checked by determining that $\det A_i \neq 0$. With enough patience these calculations could be done by hand. However, I will utilize the MATLAB Symbolic Toolbox to accelerate the process.

$$\det A_1 = \frac{9g^2(m_1 + 2m_2)}{l_1 l_2 (4m_1 + 3m_2)}$$

$$\det A_2 = -\frac{9g^2(m_1 + 2m_2)}{l_1 l_2 (4m_1 + 3m_2)}$$

$$\det A_3 = -\frac{9g^2(m_1 + 2m_2)}{l_1 l_2 (4m_1 + 3m_2)}$$

$$\det A_4 = \frac{9g^2(m_1 + 2m_2)}{l_1 l_2 (4m_1 + 3m_2)}$$

All parameters are strictly greater than zero. Thus, all equilibria are isolated. To analyze the eigenvalues of these systems, we can see if we can get any insights on the eigenvalues of matrices of the following form:

$$M = \begin{bmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ a & b & c & d \\ f & g & h & k \end{bmatrix}$$

It can be determined that the characteristic equation is:

$$\lambda^4 + (-c - k)\lambda^3 + (-a - g - dh + ck)\lambda^2 + (ak - bh + cg - df)\lambda + (ag - bf) = 0$$

We can also express each linearization as a block matrix, and then express the characteristic equation using relevant operations of the blocks.

$$A_i = \begin{bmatrix} 0 & I \\ M_K & M_B \end{bmatrix}$$

$$\lambda^4 - \text{tr}(M_B)\lambda^3 + (\det M_B - \text{tr} M_K)\lambda^2 + (\text{tr} M_K \text{tr} M_B - \text{tr} M_K M_B)\lambda + \det M_K$$

We can use the Routh-Hurwitz criterion to determine if all of the system's poles lie in the left-hand plane. They do so if all of the following are satisfied:

$$\begin{aligned}
\text{RH}_1 &= -\text{tr}M_B &> 0 \\
\text{RH}_2 &= -\text{tr}M_B(\det M_B - \text{tr}M_K) - (\text{tr}M_K\text{tr}M_B - \text{tr}M_K M_B) &> 0 \\
\text{RH}_3 &= -\text{tr}M_B(\det M_B - \text{tr}M_K)(\text{tr}M_K\text{tr}M_B - \text{tr}M_K M_B) - (\text{tr}M_K\text{tr}M_B - \text{tr}M_K M_B)^2 - \det M_K &> 0 \\
\text{RH}_4 &= \det M_K &> 0
\end{aligned}$$

Once again utilizing MATLAB, we can start with A_1 , going from the simplest inequality to the most complex.

$$\begin{aligned}
0 &\stackrel{?}{<} -\text{tr}M_B \\
&< \frac{12(b_2 l_1^2 m_1 + b_1 l_2^2 m_2 + 3b_2 l_1^2 m_2 + b_2 l_2^2 m_2 + 3b_2 l_1 l_2 m_2)}{l_1^2 l_2^2 m_2 (4m_1 + 3m_2)} \\
0 &\stackrel{?}{<} \det M_K \\
&< \frac{9g^2(m_1 + 2m_2)}{l_1 l_2 (4m_1 + 3m_2)} \\
0 &\stackrel{?}{<} -\text{tr}M_B(\det M_B - \text{tr}M_K) - (\text{tr}M_K\text{tr}M_B - \text{tr}M_K M_B) \\
&\stackrel{?}{<} -\zeta_1 b_1 g - \zeta_2 b_2 g + \zeta_3 b_1^2 b_2 + \zeta_4 b_1 b_2^2, \quad \zeta_i = h(m_1, m_2, l_1, l_2)
\end{aligned}$$

The truth of this inequality depends on the relative scale of gravity, g , and the viscous damping coefficients, b_i . In a typical real system $g \gg b_i$. If we restrict ourselves to systems like these, then the inequality does not hold.

This result implies that there exists an eigenvalue(s) of A_1 such that its real component is greater than 0. As such, by Fact 6.1 this linearization is not globally exponentially stable. Moving onto A_2 :

$$\begin{aligned}
0 &\stackrel{?}{<} -\text{tr}M_B \\
&< \frac{12(b_2 l_1^2 m_1 + b_1 l_2^2 m_2 + 3b_2 l_1^2 m_2 + b_2 l_2^2 m_2 + 3b_2 l_1 l_2 m_2)}{l_1^2 l_2^2 m_2 (4m_1 + 3m_2)} \\
0 &\stackrel{?}{<} \det M_K \\
&\not< \frac{9g^2(m_1 + 2m_2)}{l_1 l_2 (4m_1 + 3m_2)}
\end{aligned}$$

This is a nice result — no other conditions are necessary to conclude that A_2 is not exponentially stable. Now A_3 :

$$\begin{aligned}
0 &\stackrel{?}{<} \det M_K \\
&\not< \frac{9g^2(m_1 + 2m_2)}{l_1 l_2 (4m_1 + 3m_2)}
\end{aligned}$$

As with A_2 , A_3 will never be exponentially stable. Finally, A_4 . Intuitively, this equilibria would be exponentially stable for a physical system. However, trying to prove it analytically with Routh-Hurwitz is a huge pain, as we must demonstrate that all four inequalities hold. We can demonstrate positive results for the first, second, and fourth inequalities, where the left expressions always produce positive, non-zero quantities. The third term is a bit more tricky, but we can try to muscle through the reasoning.

The raw output is the following expression:

$$\begin{aligned}
& (9 * g^2 * (324 * b1^2 * l1^2 * l2^4 * m2^4 + 144 * b1^2 * l1 * l2^5 * m1 * m2^3 + 288 * b1^2 * l1 * l2^5 * m2^4 + \\
& 144 * b1 * b2 * l1^4 * l2^2 * m1^2 * m2^2 + 864 * b1 * b2 * l1^4 * l2^2 * m1 * m2^3 + 1296 * b1 * b2 * l1^4 * l2^2 * \\
& m2^4 + 1080 * b1 * b2 * l1^3 * l2^3 * m1 * m2^3 + 2592 * b1 * b2 * l1^3 * l2^3 * m2^4 + 144 * b1 * b2 * l1^2 * l2^4 * \\
& m1^2 * m2^2 + 1008 * b1 * b2 * l1^2 * l2^4 * m1 * m2^3 + 2088 * b1 * b2 * l1^2 * l2^4 * m2^4 + 288 * b1 * b2 * \\
& l1 * l2^5 * m1 * m2^3 + 576 * b1 * b2 * l1 * l2^5 * m2^4 + 144 * b2^2 * l1^5 * l2 * m1^3 * m2 + 1152 * b2^2 * l1^5 * \\
& l2 * m1^2 * m2^2 + 3024 * b2^2 * l1^5 * l2 * m1 * m2^3 + 2592 * b2^2 * l1^5 * l2 * m2^4 + 900 * b2^2 * l1^4 * l2^2 * \\
& m1^2 * m2^2 + 4320 * b2^2 * l1^4 * l2^2 * m1 * m2^3 + 5184 * b2^2 * l1^4 * l2^2 * m2^4 + 432 * b2^2 * l1^3 * l2^3 * \\
& m1^2 * m2^2 + 2808 * b2^2 * l1^3 * l2^3 * m1 * m2^3 + 4320 * b2^2 * l1^3 * l2^3 * m2^4 + 144 * b2^2 * l1^2 * l2^4 * \\
& m1^2 * m2^2 + 1008 * b2^2 * l1^2 * l2^4 * m1 * m2^3 + 1764 * b2^2 * l1^2 * l2^4 * m2^4 + 144 * b2^2 * l1 * l2^5 * \\
& m1 * m2^3 + 288 * b2^2 * l1 * l2^5 * m2^4 - 16 * l1^5 * l2^5 * m1^3 * m2^3 - 56 * l1^5 * l2^5 * m1^2 * m2^4 - 57 * \\
& l1^5 * l2^5 * m1 * m2^5 - 18 * l1^5 * l2^5 * m2^6)) / (l1^6 * l2^6 * m2^3 * (4 * m1 + 3 * m2)^3) + (9 * g * \\
& (864 * b1^3 * b2 * l2^3 * m2^2 + 864 * b1^2 * b2^2 * l1^2 * l2 * m1 * m2 + 2592 * b1^2 * b2^2 * l1^2 * l2 * m2^2 + \\
& 864 * b1^2 * b2^2 * l1 * l2^2 * m1 * m2 + 4320 * b1^2 * b2^2 * l1 * l2^2 * m2^2 + 1728 * b1^2 * b2^2 * l2^3 * m2^2 + \\
& 864 * b1 * b2^3 * l1^3 * m1^2 + 4320 * b1 * b2^3 * l1^3 * m1 * m2 + 5184 * b1 * b2^3 * l1^3 * m2^2 + 3456 * b1 * \\
& b2^3 * l1^2 * l2 * m1 * m2 + 7776 * b1 * b2^3 * l1^2 * l2 * m2^2 + 864 * b1 * b2^3 * l1 * l2^2 * m1 * m2 + 4320 * \\
& b1 * b2^3 * l1 * l2^2 * m2^2 + 864 * b1 * b2^3 * l2^3 * m2^2)) / (l1^6 * l2^6 * m2^3 * (4 * m1 + 3 * m2)^3)
\end{aligned}$$

Our first point of order can be the denominators denoted in **red**, which are equal and always positive. As such, we do not need to discuss them further. Looking at the coefficients in **blue**, we see that the first term is multiplied by one more g than the second, suggesting that the first term is dominating in this expression. This idea is further supported by terms inside the parentheses. The terms in **green** scale a combination of parameters that look like $b_i^4 l_i^3 m_i^2$. Under the assumption that the damping coefficients are small relative to gravity, taking them to the fourth power shrinks these terms, thus we can restrict the scope of this qualitative analysis to the first numerator.

The numerator takes a negative value when the **purple** terms are greater than the black terms. If we assume that the masses are similar, the arm lengths are similar, and the damping coefficients are similar, we can factor out $l_i^6 m_i^4$ from all terms in the first numerator. So we just need to compare the values of $\sum \alpha_i b_{1,2}^2$ to $\sum \beta_i l_{1,2}^4 m_{1,2}^2$. As a rough estimate, $\sum \alpha_i b_{1,2}^2 \approx 40000 b_{1,2}^2$ and $\sum \beta_i l_{1,2}^4 m_{1,2}^2 \approx -160 l_{1,2}^4 m_{1,2}^2$. Thus, stability depends on the truth of the following inequality:

$$\begin{aligned}
40000 b_{1,2}^2 &> 160 l_{1,2}^4 m_{1,2}^2 \\
b_{1,2} &> 0.0632 l_{1,2}^2 m_{1,2}
\end{aligned}$$

The physical interpretation of this inequality makes intuitive sense: the damping of the system must be greater than a threshold amount of its inertia for the states to reach its equilibrium point. So long as there is a sufficiently large damping, this linearization is exponentially stable about this equilibrium.

2.2 Fixed system realization

So far this analysis has been performed symbolically. Though it has helped us justify some intuition of the systems, it was very messy along the way. Personally, I don't feel quite satisfied with how messy and hand-wavey this process was¹. It will be easier to proceed with further analysis if we give numerical, physical values to these parameters and then determine what we can about the stability of our system.

¹ This is a way of saying that I put way too much time into doing this symbolically and I didn't get any clear-cut results, but I still wanted to show that put effort into this assignment even if it didn't go anywhere.

m_1	0.7121 kg
m_2	0.8901 kg
l_1	0.8 m
l_2	1 m
b_1	$0.05 \frac{\text{N}}{\text{rad/s}}$
b_2	$0.05 \frac{\text{N}}{\text{rad/s}}$
g	$9.81 \frac{\text{m}}{\text{s}^2}$

Table 3: Physical realization parameters.

These masses and lengths replicate using 1018 carbon steel rods that are 12 mm in diameter of lengths 0.8 m and 1 m. From here, we can compute the linearizations for this system and perform the same Routh-Hurwitz criterion checks for their eigenvalues.

$$A_1 = \begin{bmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 33.2272 & -17.8001 & -0.5436 & 0.3737 \\ -39.8726 & 36.0752 & 0.8208 & -0.6170 \end{bmatrix}$$

$$A_2 = \begin{bmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 33.2272 & -17.8001 & -0.5436 & 0.3737 \\ -39.8726 & 36.0752 & 0.8208 & -0.6170 \end{bmatrix}$$

$$A_3 = \begin{bmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 33.2272 & -17.8001 & -0.1359 & -0.0340 \\ 39.8726 & -36.0752 & 0.0054 & -0.2093 \end{bmatrix}$$

$$A_4 = \begin{bmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ -33.2272 & 17.8001 & -0.5436 & 0.3737 \\ 39.8726 & -36.0752 & 0.8208 & -0.6170 \end{bmatrix}$$

	RH ₁	RH ₂	RH ₃	RH ₄
A_1	1.16	-69.80	250.87	488.94
A_2	0.34	-1.57	487.99	-488.94
A_3	0.34	1.59	487.98	-488.93
A_4	1.16	69.87	251.58	488.94

Table 4: Routh-Hurwitz criteria values for fixed system.

The results from Table 4 would suggest that linearizations A_1 , A_2 , and A_3 have eigenvalues with positive real components and that A_4 has only eigenvalues that have a negative real component. This can be verified using the `eig()` command.

	λ_1	λ_2	λ_3	λ_4
A_1	-8.42	7.28	-2.84	2.81
A_2	4.76	-0.07+4.55i	-0.07-4.55i	-4.96
A_3	4.48	-4.63	-0.10+4.86i	-0.10-4.86i
A_4	-0.57+7.81i	-0.57-7.81i	-0.1+2.82i	-0.1-2.82i

Table 5: Fixed system linearization eigenvalues.

We can conclude that A_4 is the only linear system with $\text{Re}(\lambda) < 0$ for all eigenvalues. So by Fact 6.1, only A_4 represents a globally exponentially stable system.

We can use Theorem 9.4/Theorem 8.3 and show that the exponential stability of $\dot{x} = A_4x$ means that the equilibrium x_4^* of $\dot{x} = f(x)$ is exponentially stable.

3 Lyapunov construction

The double inverted pendulum problem and serial-linkage robotic system fundamentally have the same passive structure — with no input, they have the same dynamics. As such, we can try applying the methods in Lecture 10 to our system to see if we can construct a Lyapunov function.

$$q = \begin{bmatrix} \theta_1 \\ \theta_2 \end{bmatrix}, \quad \dot{q} = \begin{bmatrix} \dot{\theta}_1 \\ \dot{\theta}_2 \end{bmatrix}$$

$$D(q) = \begin{bmatrix} M_1 & M_3 \cos \Delta \\ M_3 \cos \Delta & M_2 \end{bmatrix}, \quad P(q) = K_2 \cos \theta_1 + K_4 \cos \theta_2$$

For our stable equilibrium point x_4^* , $P(x_4^*) = -K_2 - K_4$. We can plug in our terms into Lecture Equation (10.13) with $x_4^* = 0$:

$$\begin{aligned} V(x) &= \frac{1}{2} \dot{q}^T D(q) \dot{q} + P(q) - P(x^*) \\ &= \frac{1}{2} M_1 \dot{\theta}_1^2 + M_3 \dot{\theta}_1 \dot{\theta}_2 \cos^2 \Delta + \frac{1}{2} M_2 \dot{\theta}_2^2 - K_2 \cos \theta_1 - K_4 \cos \theta_2 + K_2 + K_4 \\ &\geq 0, \quad \forall x \in E \\ V(0) &= 0 \end{aligned}$$

$$\begin{aligned} \dot{V}(x) &= \dot{q}^T D \ddot{q} + \frac{1}{2} \dot{q}^T \dot{D} \dot{q} + \frac{\partial P}{\partial q} \dot{q} \\ &= \dot{q}^T \begin{bmatrix} -K_1 \dot{\theta}_2^2 \sin \Delta + K_2 \sin \theta_1 - B_1 \dot{\theta}_1 + B_2 \dot{\theta}_2 \\ K_3 \dot{\theta}_1^2 \sin \Delta + K_4 \sin \theta_2 + B_2 \dot{\theta}_1 - B_2 \dot{\theta}_2 \end{bmatrix} \\ &\quad + \frac{1}{2} \dot{q}^T \begin{bmatrix} 0 & -M_3(\dot{\theta}_1 - \dot{\theta}_2) \sin \Delta \\ -M_3(\dot{\theta}_1 - \dot{\theta}_2) \sin \Delta & 0 \end{bmatrix} \dot{q} \\ &\quad + \begin{bmatrix} -K_2 \dot{\theta}_1 \sin \theta_1 & -K_4 \dot{\theta}_2 \sin \theta_2 \end{bmatrix} \dot{q} \\ &= -B_1 \dot{\theta}_1^2 + 2B_2 \dot{\theta}_1 \dot{\theta}_2 - B_2 \dot{\theta}_2^2 \\ &= -b_1 \dot{\theta}_1^2 - b_2 (\dot{\theta}_1^2 - 2\dot{\theta}_1 \dot{\theta}_2 + \dot{\theta}_2^2) \\ &= -b_1 \dot{\theta}_1^2 - b_2 (\dot{\theta}_1 - \dot{\theta}_2)^2 \\ &\leq 0, \quad \forall x \in E \end{aligned}$$

From Theorem 7.1 we can conclude that for a general system, this equilibrium point is stable. We cannot conclude asymptotic stability using this theorem, as the value of $V(x) = 0$ for every $\theta_i = 2\pi n$, $n \in \mathbb{Z}$ with $x_4^* = 0$.

We cannot construct an energy-based Lyapunov function for the other equilibria of the system. The terms associated with the potential energy of the system are such that forcing $V(0) = 0$ means there exists states such that $V(x) < 0$.

In order to establish another sense of stability, we turn to Theorem 8.3. We have established that the linearization $\dot{x} = Df(0)x$ is exponentially stable. Therefore, the unique solution $P = P^T > 0$ of the CTLE:

$$Df(0)^T P + P Df(0) = -Q$$

for a given $Q = Q^T > 0$ will produce a Lyapunov function for the nonlinear system in the form $V(x) = x^T P x$. We know from Theorem 8.2 that the global exponential stability of this isolated equilibrium means that there always exists such a solution to P regardless of the choice of Q . So, we can just pick an arbitrary Q to continue the analysis. For simplicity's sake, we will set $Q = I$ and compute the solution P through the MATLAB `lyap()` function, from which we obtain:

$$P = \begin{bmatrix} 169.9562 & 78.9822 & 0.3332 & 0.2651 \\ 78.9822 & 70.3429 & -0.2273 & -0.0983 \\ 0.3332 & -0.2273 & 18.9870 & 11.5591 \\ 0.2651 & -0.0983 & 11.5591 & 7.6527 \end{bmatrix}$$

To determine the size of the ball for which the Lyapunov function $V(x) = x^T P x$ validates exponential stability, we check the inequality:

$$\max_{\|x\| \leq \rho} \left\| \int_0^1 Df(\tau x) d\tau - Df(0) \right\| < \frac{\lambda_{\min}(Q)}{2\lambda_{\max}(P)}$$

The right side can be easily computed numerically.

$$\frac{\lambda_{\min}(Q)}{2\lambda_{\max}(P)} = 0.0023$$

The left side sucks. Our Jacobian matrix is really complex, with many products and sums of sinusoidal functions. As a result, integrating them is a real headache. I have attempted to do this symbolically within MATLAB. However, as seen in Figure 3 the integral is still not resolved.

Figure 3: Attempt at determining N for Theorem 8.3

Because these are not able to be resolved with the current means in a reasonable timeframe, determining the conditions to determine the radius of the ball. However, at the very least we can say that this ball exists, and that the equilibrium is exponentially stable within it.

Let's return to Theorem 7.1. We can get asymptotic stability if $V(x) < 0 \forall x \in E \setminus \{0\}$. The issue arises from the fact that $\theta_1, \theta_2 \in \mathbb{R}$, from which every full rotation of the equilibrium is also causes the Lyapunov function to be zero. Something that could resolve this is if we could wrap such that $\theta_i \in [-\pi, \pi)$. In this case, it is true that $x^* = 0$ is the

only state that makes V be zero. If this is the case, then we can conclude asymptotic stability of this equilibrium.

Homework 3

This project will be evaluating the following double inverted pendulum system:

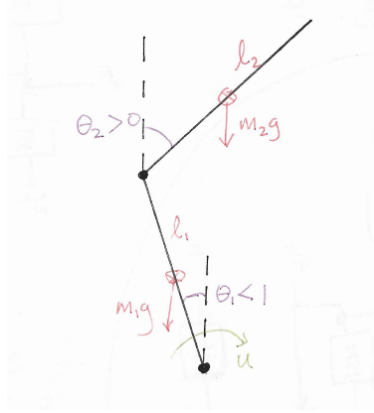


Figure 1: Double inverted pendulum.

An input torque u is applied to the first link of the pendulum. Each arm is modeled as a uniform rod ($I_i = \frac{1}{12}m_i l_i^2$) and friction between the rods is assumed as $|F_1| = b_1|\dot{\theta}_1|$ and $|F_2| = b_2|\dot{\theta}_1 - \dot{\theta}_2|$. Using the angular deviation from the vertical and angular velocities of the links, we can define the state $x = [\theta_1 \ \theta_2 \ \dot{\theta}_1 \ \dot{\theta}_2]^\top$. The dynamics of the system are driven by the following differential equation in the form $\dot{x} = f(x, u)$:

$$f(x, u) = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & \mu_2(x) & \mu_3(x) \\ 0 & 0 & \mu_3(x) & \mu_1(x) \end{bmatrix} \left(\begin{bmatrix} \nu_1(x) \\ \nu_2(x) \\ \nu_3(x) \\ \nu_4(x) \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ 1 \\ 0 \end{bmatrix} u \right) = \begin{bmatrix} \nu_1(x) \\ \nu_2(x) \\ \mu_2(x)\nu_3(x) + \mu_3(x)\nu_4(x) \\ \mu_3(x)\nu_3(x) + \mu_1(x)\nu_4(x) \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ \mu_2(x) \\ \mu_3(x) \end{bmatrix} u$$

1 Barrier function for a constant input

A method of constructing a barrier function is outlined by Proposition 12.1, from which the barrier function establishes forward invariance of the a level set, Ω_c of a satisfactory Lyapunov function, with respect to the nonlinear system $\dot{x} = f(x)$. By Definition 12.1, this forward invariance property means that the system $\dot{x} = f(x)$ is safe with respect to Ω_c — in other words Ω_c is a safety set of $\dot{x} = f(x)$.

Thus, the strategy for this assignment is to find a Lyapunov function that also accounts for the input, define the sublevel set $|\Omega_{\text{c}}|$, and use the two to create a barrier function.

We can attempt to use the same method as before to create a Lyapunov function, now with u being a non-zero control input.

$$q = \begin{bmatrix} \theta_1 \\ \theta_2 \end{bmatrix}, \quad \dot{q} = \begin{bmatrix} \dot{\theta}_1 \\ \dot{\theta}_2 \end{bmatrix}$$

$$D(q) = \begin{bmatrix} M_1 & M_3 \cos \Delta \\ M_3 \cos \Delta & M_2 \end{bmatrix}, \quad P(q) = K_2 \cos \theta_1 + K_4 \cos \theta_2$$

$$D\ddot{q} = \begin{bmatrix} -K_1\dot{\theta}_2^2 \sin \Delta + K_2 \sin \theta_1 - B_1\dot{\theta}_1 + B_2\dot{\theta}_2 + u \\ K_3\dot{\theta}_1^2 \sin \Delta + K_4 \sin \theta_2 + B_2\dot{\theta}_1 - B_2\dot{\theta}_2 \end{bmatrix}$$

As before, we will use an energy-based construction about the down-down equilibrium point such that $x^* = 0$.

$$\begin{aligned} V(x) &= \frac{1}{2} \dot{q}^T D(q) \dot{q} + P(q) - P(x^*) \\ &= \frac{1}{2} M_1 \dot{\theta}_1^2 + M_3^2 \dot{\theta}_1 \dot{\theta}_2 \cos^2 \Delta + \frac{1}{2} M_2 \dot{\theta}_2^2 - K_2 \cos \theta_1 - K_4 \cos \theta_2 + K_2 + K_4 \end{aligned}$$

$$\begin{aligned} \dot{V}(x) &= \dot{q}^T D\ddot{q} + \frac{1}{2} \dot{q}^T \dot{D}\dot{q} + \frac{\partial P}{\partial q} \dot{q} \\ &= \dot{q}^T \begin{bmatrix} -K_1\dot{\theta}_2^2 \sin \Delta + K_2 \sin \theta_1 - B_1\dot{\theta}_1 + B_2\dot{\theta}_2 + u \\ K_3\dot{\theta}_1^2 \sin \Delta + K_4 \sin \theta_2 + B_2\dot{\theta}_1 - B_2\dot{\theta}_2 \end{bmatrix} \\ &\quad + \frac{1}{2} \dot{q}^T \begin{bmatrix} 0 & -M_3(\dot{\theta}_1 - \dot{\theta}_2) \sin \Delta \\ -M_3(\dot{\theta}_1 - \dot{\theta}_2) \sin \Delta & 0 \end{bmatrix} \dot{q} \\ &\quad + \begin{bmatrix} -K_2 \sin \theta_1 & -K_4 \sin \theta_2 \end{bmatrix} \dot{q} \\ &= -b_1 \dot{\theta}_1^2 - b_2 (\dot{\theta}_1 - \dot{\theta}_2)^2 + \dot{\theta}_1 u \end{aligned}$$

There are a couple things wrong with this current function. The first is that there exists $x \in E \setminus \{0\}$ such that $V(x) = 0$ due to the wrapping of θ_1 and θ_2 . However, this can be resolved if we reduce the size of E such that $\theta_1, \theta_2 \in (-2\pi, 2\pi)$. The second is the failure of $\dot{V}(x) \leq 0$. The appearance of $\dot{\theta}_1 u$ means that there exists states for which $\dot{V}(x) > 0$.

While both of these issues need correction, there is also a more fundamental level of understanding that this Lyapunov function fails to consider: the shifting of our equilibrium point. From the physical understanding of the problem, by applying a constant input torque, our best case is a shifted equilibrium point and our worst case is a torque large enough to cause constant rotation of our pendulum. Thus, we need to return to the equilibrium problem.

2 System with non-zero constant u

2.1 Revisiting equilibrium

We will be discovering the set of equilibrium points for our system. Because u is a constant, we will incorporate it into our dynamics such that we still have a system in the form $\dot{x} = f(x)$.

$$\begin{aligned}
f(x) &= \begin{bmatrix} \nu_1(x) \\ \nu_2(x) \\ \mu_2(x)\nu_3(x) + \mu_3(x)\nu_4(x) + \mu_2(x)u \\ \mu_3(x)\nu_3(x) + \mu_1(x)\nu_4(x) + \mu_3(x)u \end{bmatrix} = 0 \\
&\Rightarrow \nu_1(x) = \dot{\theta}_1 = 0 \\
&\Rightarrow \nu_2(x) = \dot{\theta}_2 = 0 \\
\mu_2(x)\nu_3(x) + \mu_3(x)\nu_4(x) + \mu_2(x)u \Big|_{\dot{\theta}_1, \dot{\theta}_2=0} &= \frac{M_2 K_2 \sin \theta_1 - M_3 K_4 \cos \Delta \sin \theta_2 + M_2 u}{|M(x)|} = 0 \\
\mu_3(x)\nu_3(x) + \mu_1(x)\nu_4(x) + \mu_3(x)u \Big|_{\dot{\theta}_1, \dot{\theta}_2=0} &= \frac{-M_3 K_2 \cos \Delta \sin \theta_1 + M_1 K_4 \sin \theta_2 - M_3 u \cos \Delta}{|M(x)|} = 0 \\
\begin{bmatrix} M_2 K_2 & -M_3 K_4 \cos \Delta \\ -M_3 K_2 \cos \Delta & M_1 K_4 \end{bmatrix} \begin{bmatrix} \sin \theta_1 \\ \sin \theta_2 \end{bmatrix} &= \begin{bmatrix} -M_2 u \\ M_3 u \cos \Delta \end{bmatrix} \\
\begin{bmatrix} \sin \theta_1 \\ \sin \theta_2 \end{bmatrix} &= \frac{1}{K_2 K_4 \cdot |M(x)|} \begin{bmatrix} M_1 K_4 & M_3 K_4 \cos \Delta \\ M_3 K_2 \cos \Delta & M_2 K_2 \end{bmatrix} \begin{bmatrix} -M_2 u \\ M_3 u \cos \Delta \end{bmatrix} \\
\Rightarrow \sin \theta_1 &= \frac{-(M_1 M_2 - M_3^2 \cos^2 \Delta)}{K_2 K_4 \cdot |M(x)|} K_4 u \\
&= -\frac{|M(x)|}{K_2 K_4 \cdot |M(x)|} K_4 u \\
&= -\frac{1}{K_2} u \\
\Rightarrow \sin \theta_2 &= 0
\end{aligned}$$

This is so cool. Our condition for θ_2 is the same, so any equilibrium points will have $\theta_2 = n\pi$, $n \in \mathbb{Z}$. The other equation forces a condition on u for equilibria to exist: $|u| \leq K_2 = (\frac{1}{2}m_1 + m_2)gl_1$. When the condition on u is met, two solutions exist for $\theta_1 \in [0, 2\pi)$, representing a solution $\sin^{-1}\left(-\frac{u}{K_2}\right) \in [-\frac{\pi}{2}, \frac{\pi}{2}]$, and its reflection across $\pi - \sin^{-1}\left(-\frac{u}{K_2}\right)$. For ease of representation, we will let $u = \gamma K_2$ where $\gamma \in [-1, 1]$. On a larger interval of E , there are more solutions every $2\pi n$, $n \in \mathbb{Z}$ revolutions of these solutions. Because these solutions are equivalent to their counter parts in $\theta_1 \in [0, 2\pi)$, only need to examine the stability of four cases.

$$x_1^* = \begin{bmatrix} \sin^{-1}(-\gamma) \\ 0 \\ 0 \\ 0 \end{bmatrix}, x_2^* = \begin{bmatrix} \pi - \sin^{-1}(-\gamma) \\ 0 \\ 0 \\ 0 \end{bmatrix}, x_3^* = \begin{bmatrix} \sin^{-1}(-\gamma) \\ \pi \\ 0 \\ 0 \end{bmatrix}, x_4^* = \begin{bmatrix} \pi - \sin^{-1}(-\gamma) \\ \pi \\ 0 \\ 0 \end{bmatrix}$$

2.2 Revisiting stability

We will use the same parameters as before for a fixed realization of the system to determine the stability of each equilibrium point. In addition, we will apply an input torque of $u = 0.5K_2$ to the system.

m_1	0.7121 kg
m_2	0.8901 kg
l_1	0.8 m
l_2	1 m
b_1	$0.05 \frac{\text{N}}{\text{rad/s}}$
b_2	$0.05 \frac{\text{N}}{\text{rad/s}}$
g	$9.81 \frac{\text{m}}{\text{s}^2}$

Table 1: Physical realization parameters.

$$A_i = Df(x_i^*, u) \Big|_{x_i^*, u} = \begin{bmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ Df_{31}(x_i^*, u) & Df_{32}(x_i^*, u) & Df_{33}(x_i^*, u) & Df_{34}(x_i^*, u) \\ Df_{41}(x_i^*, u) & Df_{42}(x_i^*, u) & Df_{43}(x_i^*, u) & Df_{44}(x_i^*, u) \end{bmatrix}$$

$$Df_{3j}(x_i^*, u) = \frac{\partial \mu_2(x_i^*)}{\partial \star_j} (\nu_3(x_i^*) + u) + \mu_2(x_i^*) \frac{\partial \nu_3(x_i^*)}{\partial \star_j} + \frac{\partial \mu_3(x_i^*)}{\partial \star_j} \nu_4(x_i^*) + \mu_3(x_i^*) \frac{\partial \nu_4(x_i^*)}{\partial \star_j}$$

$$Df_{4j}(x_i^*, u) = \frac{\partial \mu_3(x_i^*)}{\partial \star_j} (\nu_3(x_i^*) + u) + \mu_3(x_i^*) \frac{\partial \nu_3(x_i^*)}{\partial \star_j} + \frac{\partial \mu_1(x_i^*)}{\partial \star_j} \nu_4(x_i^*) + \mu_1(x_i^*) \frac{\partial \nu_4(x_i^*)}{\partial \star_j}$$

$$A_1 = \begin{bmatrix} 0 & 0 & 1.0000 & 0 \\ 0 & 0 & 0 & 1.0000 \\ 21.1135 & -11.3107 & -0.3788 & 0.2542 \\ -21.9418 & 26.4695 & 0.5622 & -0.4327 \end{bmatrix}$$

$$A_2 = \begin{bmatrix} 0 & 0 & 1.0000 & 0 \\ 0 & 0 & 0 & 1.0000 \\ -21.1135 & 11.3107 & -0.1198 & -0.0049 \\ -21.9418 & 26.4695 & 0.0441 & -0.1736 \end{bmatrix}$$

$$A_3 = \begin{bmatrix} 0 & 0 & 1.0000 & 0 \\ 0 & 0 & 0 & 1.0000 \\ 21.1135 & -11.3107 & -0.1198 & -0.0049 \\ 21.9418 & -26.4695 & 0.0441 & -0.1736 \end{bmatrix}$$

$$A_4 = \begin{bmatrix} 0 & 0 & 1.0000 & 0 \\ 0 & 0 & 0 & 1.0000 \\ -21.1135 & 11.3107 & -0.3788 & 0.2542 \\ 21.9418 & -26.4695 & 0.5622 & -0.4327 \end{bmatrix}$$

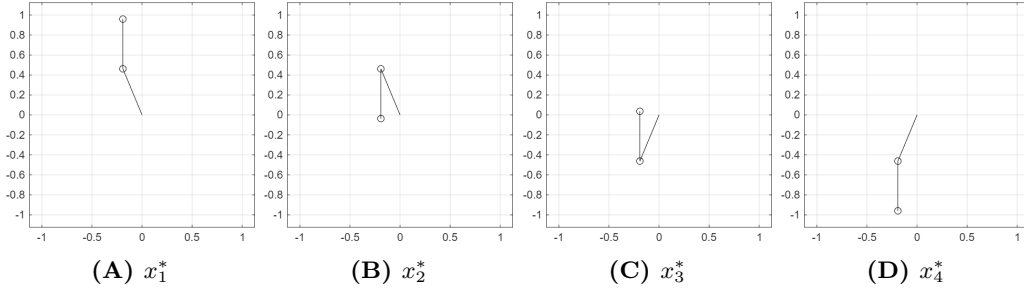
	RH ₁	RH ₂	RH ₃	RH ₄
A ₁	0.8115	-47.56	-7.226	310.7
A ₂	0.2934	-5.3349	-0.1102	-310.7
A ₃	0.2934	5.377	0.1102	-310.7
A ₄	0.8115	47.60	7.226	310.7

Table 2: Routh-Hurwitz criterion for linearized system equilibria.

	λ_1	λ_2	λ_3	λ_4
A ₁	-6.71	-2.81	5.93	2.78
A ₂	4.45	-0.6+3.89i	-0.6-3.89i	-4.61
A ₃	3.83	-3.96	-0.08+4.53i	-0.08-4.53i
A ₄	-0.39+6.29i	-0.39-6.29i	-0.1+2.80i	-0.1-2.80i

Table 3: Eigenvalues of linearized system equilibria.

We can determine that x_4^* is globally exponentially stable for its linearized system (Fact 6.1), thus the original nonlinear system is also exponentially stable for this equilibrium (Theorem 9.4).

**Figure 2:** System equilibria of interest.

2.3 Lyapunov construction

With our new equilibrium point established, we can try the same formulation for a Lyapunov function with $x_4^* = 0$.

$$\begin{aligned}
 P(x_4^*) &= K_2 \cos(\pi - \sin^{-1}(-\gamma)) - K_4 \\
 &= -K_2 \cos(\sin^{-1}(-\gamma)) - K_4 \\
 &= -K_2 \sqrt{1 - \gamma^2} - K_4
 \end{aligned}$$

$$\begin{aligned}
\mathbf{V}(x) &= \frac{1}{2} \dot{q}^T D(q) \dot{q} + P(q) - P(q^*) \\
&= \frac{1}{2} M_1 \dot{\theta}_1^2 + M_3^2 \dot{\theta}_1 \dot{\theta}_2 \cos^2 \Delta + \frac{1}{2} M_2 \dot{\theta}_2^2 + K_2 \cos(\theta_1 + \theta_1^*) - K_4 \cos \theta_2 + K_2 \sqrt{1 - \gamma^2} + K_4
\end{aligned}$$

$$\begin{aligned}
\dot{\mathbf{V}}(x) &= \dot{q}^T D \ddot{q} + \frac{1}{2} \dot{q}^T \dot{D} \dot{q} + \frac{\partial P}{\partial q} \dot{q} \\
&= -b_1 \dot{\theta}_1^2 - b_2 (\dot{\theta}_1 - \dot{\theta}_2)^2 + \dot{\theta}_1 u
\end{aligned}$$

We are now back to a similar position as before. The first thing we must do is limit θ_2 to the interval $(-2\pi, 2\pi)$. However, we still fail $\dot{\mathbf{V}}(x) \leq 0$ and $\mathbf{V}(x) > 0$, $x \in E \setminus \{0\}$. We can resolve $\dot{\mathbf{V}}(x) \leq 0$ by inserting a $-\theta_1 u$ term to the $\mathbf{V}(x)$. Finding an analytic solution to the general system requires solving this inequality of troublesome components:

$$\begin{aligned}
0 &\leq K_2 \cos(\theta_1 + \theta_1^*) + K_2 \sqrt{1 - \gamma^2} - \theta_1 u, \quad \theta_1 \neq 0 \\
&= K_2 (\cos \theta_1 \cos \theta_1^* - \sin \theta_1 \sin \theta_1^*) + K_2 \sqrt{1 - \gamma^2} - K_2 \theta_1 \gamma \\
&= K_2 \left(-\sqrt{1 - \gamma^2} \cos \theta_1 + \gamma \sin \theta_1 \right) + K_2 \sqrt{1 - \gamma^2} - K_2 \theta_1 \gamma \\
&= -K_2 \sqrt{1 - \gamma^2} \cos \theta_1 + K_2 \sqrt{1 - \gamma^2} + K_2 \gamma \sin \theta_1 - K_2 \theta_1 \gamma \\
&= K_2 \sqrt{1 - \gamma^2} (1 - \cos \theta_1) + K_2 \gamma (\sin \theta_1 - \theta_1) \\
0 &\leq \sqrt{1 - \gamma^2} (1 - \cos \theta_1) + \gamma (\sin \theta_1 - \theta_1)
\end{aligned}$$

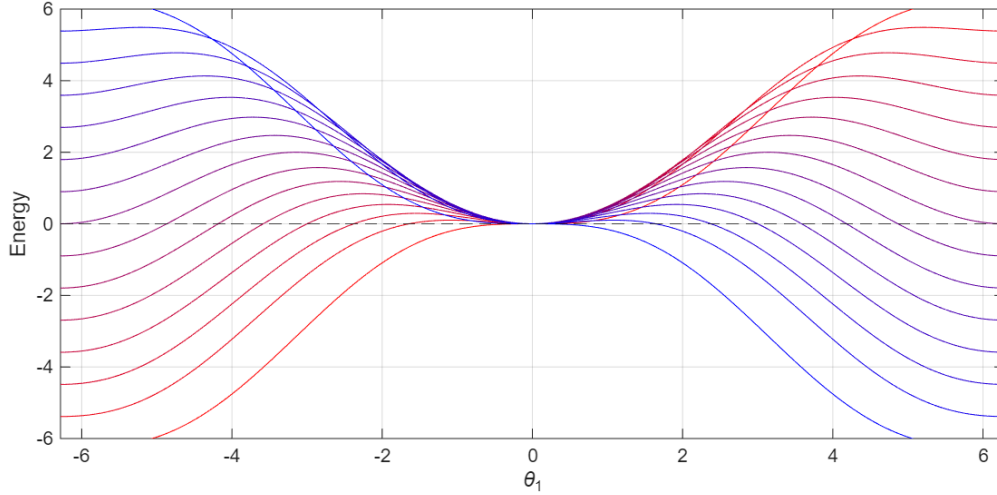


Figure 3: Energy of troublesome components of Lyapunov function. Each realization is associated with a unique $\gamma \in [-1, 1]$, with red being $\gamma = -1$ and blue being $\gamma = 1$.

This is a troublesome inequality. For any non-zero γ , the inequality is only true for an interval of θ_1 . Moreover, the size of this interval depends on the value of γ itself, which can be seen in Figure 3.

3 Fixed realization

For the purpose of this assignment, I will use the physical realization of the system and fix $\gamma = 0.4$. The inequality becomes

$$0.4 \sin \theta_1 - 0.9165 \cos \theta_1 - 0.4\theta_1 + 0.9165 \geq 0$$

With solving numerically with MATLAB via the `vpasolve()` function, we determine that this the left side has a value of zero at $\theta_1 = 0$ and $\theta_1 = 3.7015$. Thus, we will limit our set of states such that $\theta_1 \in (-\infty, 3.7015)$.

So to fully capture our findings, with our state $x = [\theta_1 \quad \theta_2 \quad \dot{\theta}_1 \quad \dot{\theta}_2]^T$, our nonlinear system is $\dot{x} = f(x, u)$ with $f : E \times [-K_2, K_2] \mapsto \mathbb{R}^4$ where $E : (-\infty, 3.7015) \times (-2\pi, 2\pi) \times \mathbb{R}^2 \subseteq \mathbb{R}^4$ and $K_2 = (\frac{1}{2}m_1 + m_2)gl_1$. If we fix our input u to be $u = 0.4K_2$, on the set E we have the following Lyapunov function around the stable equilibrium point $x^* = 0$:

$$\begin{aligned} V(x) &= \frac{1}{2} \dot{q}^T D(q) \dot{q} + P(q) - P(q^*) - \theta_1 u \\ &= \frac{1}{2} M_1 \dot{\theta}_1^2 + M_3^2 \dot{\theta}_1 \dot{\theta}_2 \cos^2 \Delta + \frac{1}{2} M_2 \dot{\theta}_2^2 \\ &\quad + K_2 \cos(\theta_1 + \theta_1^*) - K_4 \cos \theta_2 + K_2 \sqrt{1 - \gamma^2} + K_4 \\ &> 0, \forall x \in E \setminus \{0\} \\ V(0) &= 0 \end{aligned}$$

$$\begin{aligned} \dot{V}(x) &= \dot{q}^T D \ddot{q} + \frac{1}{2} \dot{q}^T \dot{D} \dot{q} + \frac{\partial P}{\partial q} \dot{q} - \dot{\theta}_1 u \\ &= -b_1 \dot{\theta}_1^2 - b_2 (\dot{\theta}_1 - \dot{\theta}_2)^2 + \dot{\theta}_1 u - \dot{\theta}_1 u \\ &= -b_1 \dot{\theta}_1^2 - b_2 (\dot{\theta}_1 - \dot{\theta}_2)^2 \\ &\leq 0, \forall x \in E \end{aligned}$$

Thus, this is a valid Lyapunov function for our nonlinear system.

3.1 Safety set

To generate a safety set, we need to establish forward invariance of our set of interest. A useful statement is Proposition 7.1. All elements of this proposition are satisfied (i.e. local Lipschitz continuity, $x^* = 0 \in E$). Thus, all sublevel sets, Ω_c , of our Lyapunov function are forward invariant for all $c > 0$, so long as $\Omega_c \subset E$.

Dy Definition 12.1, our system $\dot{x} = f(x)$ is safe with respect to any $\Omega_c \in E$. Let us choose $c = 1.7$. We can define a safety set $\mathcal{S}$ through a Lyapunov sublevel set definition

$$\mathcal{S} = \{x \in E \subseteq \mathbb{R}^4 : V(x) \leq 1.7\}$$

By Preposition 12.1, we construct a barrier function by $h(x) = 1.7 - V(x)$, which establishes forward invariance of the Lyapunov sublevel set $\Omega_{1.7} = \mathcal{S}$, our safe set.

Appendix

3.2 System equations

$$\begin{aligned}\mu_2(x) &= \frac{M_2}{|M(x)|}, \quad \mu_3(x) = \frac{-M_3 \cos \Delta}{|M(x)|}, \quad \mu_1(x) = \frac{M_1}{|M(x)|} \\ \nu_1(x) &= \dot{\theta}_1, \quad \nu_2(x) = \dot{\theta}_2 \\ \nu_3(x) &= -K_1 \dot{\theta}_2^2 \sin \Delta + K_2 \sin \theta_1 - B_1 \dot{\theta}_1 + B_2 \dot{\theta}_2 \\ \nu_4(x) &= K_3 \dot{\theta}_1^2 \sin \Delta + K_4 \sin \theta_2 + B_2 \dot{\theta}_1 - B_2 \dot{\theta}_2\end{aligned}$$

with parameters defined in Table 4.

M_1	$(\frac{1}{3}m_1 + m_2) l_1^2$
M_2	$\frac{1}{3}m_2 l_2^2$
M_3	$\frac{1}{2}m_2 l_1 l_2$
K_1	$\frac{1}{2}m_2 l_1 l_2$
K_2	$(\frac{1}{2}m_1 + m_2) gl_1$
K_3	$\frac{1}{2}m_2 l_1 l_2$
K_4	$\frac{1}{2}m_2 gl_2$
B_1	$(b_1 + b_2)$
B_2	b_2
Δ	$\theta_1 - \theta_2$
$ M(x) $	$M_1 M_2 - M_3^2 \cos^2 \Delta$

Table 4: Coefficients for rest of derivations.

3.3 Jacobian

The Jacobian of the system is:

$$Df(x) = \begin{bmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ Df_{31}(x) & Df_{32}(x) & Df_{33}(x) & Df_{34}(x) \\ Df_{41}(x) & Df_{42}(x) & Df_{43}(x) & Df_{44}(x) \end{bmatrix}$$

where the elements are defined by the equations below:

$$\begin{aligned}
Df_{31}(x) &= \frac{\partial \mu_2(x)}{\partial \theta_1} \nu_3(x) + \mu_2(x) \frac{\partial \nu_3(x)}{\partial \theta_1} + \frac{\partial \mu_3(x)}{\partial \theta_1} \nu_4(x) + \mu_3(x) \frac{\partial \nu_4(x)}{\partial \theta_1} \\
Df_{32}(x) &= \frac{\partial \mu_2(x)}{\partial \theta_2} \nu_3(x) + \mu_2(x) \frac{\partial \nu_3(x)}{\partial \theta_2} + \frac{\partial \mu_3(x)}{\partial \theta_2} \nu_4(x) + \mu_3(x) \frac{\partial \nu_4(x)}{\partial \theta_2} \\
Df_{33}(x) &= \mu_2(x) \frac{\partial \nu_3(x)}{\partial \dot{\theta}_1} + \mu_3(x) \frac{\partial \nu_4(x)}{\partial \dot{\theta}_1} \\
Df_{34}(x) &= \mu_2(x) \frac{\partial \nu_3(x)}{\partial \dot{\theta}_2} + \mu_3(x) \frac{\partial \nu_4(x)}{\partial \dot{\theta}_2} \\
Df_{41}(x) &= \frac{\partial \mu_3(x)}{\partial \theta_1} \nu_3(x) + \mu_3(x) \frac{\partial \nu_3(x)}{\partial \theta_1} + \frac{\partial \mu_1(x)}{\partial \theta_1} \nu_4(x) + \mu_1(x) \frac{\partial \nu_4(x)}{\partial \theta_1} \\
Df_{42}(x) &= \frac{\partial \mu_3(x)}{\partial \theta_2} \nu_3(x) + \mu_3(x) \frac{\partial \nu_3(x)}{\partial \theta_2} + \frac{\partial \mu_1(x)}{\partial \theta_2} \nu_4(x) + \mu_1(x) \frac{\partial \nu_4(x)}{\partial \theta_2} \\
Df_{43}(x) &= \mu_3(x) \frac{\partial \nu_3(x)}{\partial \dot{\theta}_1} + \mu_1(x) \frac{\partial \nu_4(x)}{\partial \dot{\theta}_1} \\
Df_{44}(x) &= \mu_3(x) \frac{\partial \nu_3(x)}{\partial \dot{\theta}_2} + \mu_1(x) \frac{\partial \nu_4(x)}{\partial \dot{\theta}_2}
\end{aligned}$$

$$D \begin{bmatrix} \mu_1(x) \\ \mu_2(x) \\ \mu_3(x) \end{bmatrix} = \begin{bmatrix} -M_1 M_3 \frac{\cos \Delta \sin \Delta}{|M(x)|^2} & M_1 M_3 \frac{\cos \Delta \sin \Delta}{|M(x)|^2} & 0 & 0 \\ -M_2 M_3 \frac{\cos \Delta \sin \Delta}{|M(x)|^2} & M_2 M_3 \frac{\cos \Delta \sin \Delta}{|M(x)|^2} & 0 & 0 \\ -M_3 \frac{(2M_3 \cos^2 \Delta - |M(x)|) \sin \Delta}{|M(x)|^2} & M_3 \frac{(2M_3 \cos^2 \Delta - |M(x)|) \sin \Delta}{|M(x)|^2} & 0 & 0 \end{bmatrix}$$

$$D \begin{bmatrix} \nu_1(x) \\ \nu_2(x) \\ \nu_3(x) \\ \nu_4(x) \end{bmatrix} = \begin{bmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ -K_1 \dot{\theta}_2^2 \cos \Delta + K_2 \cos \theta_1 & K_1 \dot{\theta}_2^2 \cos \Delta & -B_1 & -2K_1 \dot{\theta}_2 \sin \Delta + B_2 \\ K_3 \dot{\theta}_1^2 \cos \Delta & -K_3 \dot{\theta}_1^2 \cos \Delta + K_4 \cos \theta_2 & 2K_3 \dot{\theta}_1 \sin \Delta + B_2 & -B_2 \end{bmatrix}$$

Homework 4

1 SISO feedback linearization

Theorem 20.1 gives us the groundwork for using feedback linearization to create a controller that makes a SISO system exponentially stable (thus asymptotically stable). For this theorem to apply, we need the following:

- A SISO system with relative degree γ
- Zero dynamics $\dot{z} = \omega(0, z)$ with ω continuously differentiable
- Exponentially stable equilibrium $z^* = 0$
- Closed-loop output dynamics $\dot{\eta} = A_{cl}\eta$ with stable equilibrium $\eta^* = 0$

1.1 Normal form transformation

Our system is control affine, and can be written as

$$\dot{x} = \underbrace{\begin{bmatrix} \nu_1(x) \\ \nu_2(x) \\ \mu_2(x)\nu_3(x) + \mu_3(x)\nu_4(x) \\ \mu_3(x)\nu_3(x) + \mu_1(x)\nu_4(x) \end{bmatrix}}_{f(x)} + \underbrace{\begin{bmatrix} 0 \\ 0 \\ \mu_2(x) \\ \mu_3(x) \end{bmatrix}}_{g(x)} u$$

For our output, we will choose the angle of the first joint, which will be driven to its upright position $\theta_1 = 0$. Thus, $y = h(x) = \theta_1$. We will now take Lie derivatives to determine the relative degree γ .

$$\begin{aligned} L_f h(x) &= \begin{bmatrix} 1 & 0 & 0 & 0 \end{bmatrix} f(x) = \nu_1(x) = \dot{\theta}_1 \\ L_g h(x) &= \begin{bmatrix} 1 & 0 & 0 & 0 \end{bmatrix} g(x) = 0 \\ L_f^2 h(x) &= \begin{bmatrix} 0 & 0 & 1 & 0 \end{bmatrix} f(x) = \mu_2(x)\nu_3(x) + \mu_3(x)\nu_4(x) \\ L_g L_f h(x) &= \begin{bmatrix} 0 & 0 & 1 & 0 \end{bmatrix} g(x) = \mu_2(x) = \frac{M_2}{|M(x)|} > 0 \quad \forall x \in E \\ &\Rightarrow \gamma = 2 \end{aligned}$$

We have obtained the relative degree γ . Our next goal is to perform a coordinate transform to represent our system in normal form:

$$\begin{aligned} \dot{\eta} &= \hat{f}(\eta, z) + \hat{g}(\eta, z)u \\ \dot{z} &= \omega(\eta, z) \\ y &= \eta_1 \end{aligned}$$

A normal form transformation is described in Definition 19.1, where it takes the form

$$\begin{bmatrix} \eta \\ z \end{bmatrix} = \Phi(x) = \begin{bmatrix} h(x) \\ L_f h(x) \\ z_1(x) \\ z_2(x) \end{bmatrix}$$

where each row is linearly independent at $x \in E$ and $L_g z_i(x) \equiv 0$. An obvious first choice is $z_1(x) = \theta_2$:

$$L_g z_1(x) = \begin{bmatrix} 0 & 1 & 0 & 0 \end{bmatrix} g(x) = 0$$

For $z_2(x)$, it can be shown that $z_2(x) = \frac{M_3 \cos \Delta}{M_2} \dot{\theta}_1 + \dot{\theta}_2$ is suitable.

$$\begin{aligned} z_2(x) &= \frac{M_3 \cos \Delta}{M_2} \dot{\theta}_1 + \dot{\theta}_2 \\ L_g z_2(x) &= \begin{bmatrix} -\frac{M_3 \sin \Delta}{M_2} \dot{\theta}_1 & \frac{M_3 \sin \Delta}{M_2} \dot{\theta}_1 & \frac{M_3 \cos \Delta}{M_2} & 1 \end{bmatrix} g(x) \\ &= \frac{M_3 \cos \Delta}{M_2} \frac{M_2}{|M(x)|} - \frac{M_3 \cos \Delta}{|M(x)|} \\ &= 0 \end{aligned}$$

Thus,

$$\Phi(x) = \begin{bmatrix} \theta_1 \\ \dot{\theta}_1 \\ \theta_2 \\ \frac{M_3 \cos \Delta}{M_2} \dot{\theta}_1 + \dot{\theta}_2 \end{bmatrix}$$

is our coordinate transform, which is also a local diffeomorphism by Theorem 19.3. We can confirm this by looking at its Jacobian.

$$D\Phi(x) = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ -\frac{M_3 \sin \Delta}{M_2} \dot{\theta}_1 & \frac{M_3 \sin \Delta}{M_2} \dot{\theta}_1 & \frac{M_3 \cos \Delta}{M_2} & 1 \end{bmatrix}$$

$$\det(D\Phi(x)) = -1, \forall x \in E$$

We can also construct its inverse, $x = \Phi^{-1}(\eta, z)$:

$$\Phi^{-1}(\eta, z) = \begin{bmatrix} \eta_1 \\ z_1 \\ \eta_2 \\ z_2 - \eta_2 \left(\frac{M_3}{M_2} \cos \Delta_n \right) \end{bmatrix}$$

with $\Delta_n = \eta_1 - z_1$.

1.2 Normal and zero dynamics

By Lecture equation (19.18), the normal dynamics will be given by:

$$\dot{z} = \begin{bmatrix} Dz_1(x)f(x) \\ Dz_2(x)f(x) \end{bmatrix} = \underbrace{\begin{bmatrix} Dz_1(\Phi^{-1}(\eta, z))f(\Phi^{-1}(\eta, z)) \\ Dz_2(\Phi^{-1}(\eta, z))f(\Phi^{-1}(\eta, z)) \end{bmatrix}}_{\omega(\eta, z)}$$

$$\begin{aligned} \omega_1(\eta, z) &= Dz_1(\Phi^{-1}(\eta, z))f(\Phi^{-1}(\eta, z)) = \begin{bmatrix} 0 & 1 & 0 & 0 \end{bmatrix} \begin{bmatrix} \nu_1(x) \\ \nu_2(x) \\ \mu_2(x)\nu_3(x) + \mu_3(x)\nu_4(x) \\ \mu_3(x)\nu_3(x) + \mu_1(x)\nu_4(x) \end{bmatrix} \Big|_{x=\Phi^{-1}(\eta, z)} \\ &= \nu_2(\Phi^{-1}(\eta, z)) = z_2 - \eta_2 \left(\frac{M_3}{M_2} \cos \Delta_n \right) \end{aligned}$$

$$\begin{aligned} \omega_2(\eta, z) &= Dz_2(\Phi^{-1}(\eta, z))f(\Phi^{-1}(\eta, z)) = \begin{bmatrix} -\frac{M_3 \sin \Delta}{M_2} \dot{\theta}_1 & \frac{M_3 \sin \Delta}{M_2} \dot{\theta}_1 & \frac{M_3 \cos \Delta}{M_2} & 1 \end{bmatrix} \begin{bmatrix} \nu_1(x) \\ \nu_2(x) \\ \mu_2(x)\nu_3(x) + \mu_3(x)\nu_4(x) \\ \mu_3(x)\nu_3(x) + \mu_1(x)\nu_4(x) \end{bmatrix} \Big|_{x=\Phi^{-1}(\eta, z)} \\ &= \frac{M_3}{M_2} \eta_2 \sin \Delta_n \left(z_2 - \eta_2 \left(1 + \frac{M_3}{M_2} \cos \Delta_n \right) \right) \\ &\quad + \frac{M_3 \sin \Delta_n}{M_2} (\mu_2(\Phi^{-1}(\eta, z))\nu_3(\Phi^{-1}(\eta, z)) + \mu_3(\Phi^{-1}(\eta, z))\nu_4(\Phi^{-1}(\eta, z))) \\ &\quad + \mu_3(\Phi^{-1}(\eta, z))\nu_3(\Phi^{-1}(\eta, z)) + \mu_1(\Phi^{-1}(\eta, z))\nu_4(\Phi^{-1}(\eta, z)) \\ &= \frac{M_3 \sin \Delta}{M_2} \dot{\theta}_1 (f_2(x) - f_1(x)) + \frac{M_3 \cos \Delta}{M_2} f_3(x) + f_4(x) \Big|_{x=\Phi^{-1}(\eta, z)} \end{aligned}$$

$$D\omega(\eta, z) = \begin{bmatrix} 0 & 1 \\ *_1 & *_2 \end{bmatrix}$$

The numerators of $*_1$ and $*_2$ are products of sinusoids and polynomials, thus they will always be continuously differentiable. Their denominators will be products of constants and $(M_1 M_2 - M_3^2 \cos^2(\eta_1 - z_1))^2$. In Homework 1, we demonstrated that this value will always be non-zero for all $x \in E$. Because $\eta_1 = \theta_1$ and $z_1 = \theta_2$, this value will also be non-zero for all $(\eta, z) \in Y \times Z$. Thus, all elements in $D\omega(\eta, z)$ are continuous functions and $\omega(\eta, z)$ is continuously differentiable.

The results that we want to use rely on the zero dynamics $\dot{z} = \omega(0, z)$ and its exponentially stable equilibrium $z^* = 0$. We will take the time to find out more about these items.

$$\begin{aligned} \omega_1(0, z) &= z_2 \\ \Rightarrow z_2 &= 0 \text{ for equilibrium} \end{aligned}$$

$$\begin{aligned} \omega_2(0, z)|_{z_2=0} &= \frac{M_3}{M_2} \sin(-z_1) \left(\frac{-M_3 \cos(-z_1)}{M_1 M_2 - M_3^2 \cos^2(-z_1)} K_4 \sin(z_1) \right) + \frac{M_1}{M_1 M_2 - M_3^2 \cos^2(-z_1)} K_4 \sin(z_1) \\ &= K_4 \sin(z_1) \cdot \frac{M_1 - \frac{M_3^2}{2M_2} \sin(-2z_1)}{M_1 M_2 - M_3^2 \cos^2(-z_1)} \\ &\Rightarrow z_1 = n\pi, \quad n \in \mathbb{Z} \text{ for equilibrium} \end{aligned}$$

Our zero dynamics equilibria¹ are $z_1^* = \begin{bmatrix} 0 & 0 \end{bmatrix}^T$ and $z_2^* = \begin{bmatrix} \pi & 0 \end{bmatrix}^T$. We would like to establish stability for these equilibria. We can try examining their linearizations, $A_i = D\omega(0, z_i^*)$.

$$D\omega(0, z_1^*) = A_1 = \begin{bmatrix} 0 & 1 \\ \frac{K_4}{M_2} & -\frac{B_2}{M_2} \end{bmatrix}$$

$$D\omega(0, z_2^*) = A_2 = \begin{bmatrix} 0 & 1 \\ -\frac{K_4}{M_2} & -\frac{B_2}{M_2} \end{bmatrix}$$

These matrices look similar, thus we can see what the Routh-Hurwitz criterion looks like for a matrix of the same form.

$$A = \begin{bmatrix} 0 & 1 \\ a & b \end{bmatrix}$$

$$\lambda I - A = \begin{bmatrix} \lambda & -1 \\ -a & \lambda - b \end{bmatrix}$$

$$\det(\lambda I - A) = \lambda^2 - b\lambda - a = 0$$

By the Routh-Hurwitz criterion, the above polynomial will have eigenvalues with negative real parts when the following inequalities are true:

$$-b > 0$$

$$-a > 0$$

We can break our constants into physical parameters to determine if these inequalities hold for the linearizations. For the linearization about z_1^* :

$$-b = \frac{B_2}{M_2} > 0$$

$$-a = -\frac{K_4}{M_2} \not> 0$$

Thus, $z_1^* = \begin{bmatrix} 0 & 0 \end{bmatrix}^T$ is not an exponentially stable equilibrium point. Going to the next equilibrium:

$$-b = \frac{B_2}{M_2} > 0$$

$$-a = \frac{K_4}{M_2} > 0$$

The eigenvalues of A_2 have negative real components and the equilibrium point is isolated, so A_2 is exponentially stable (Fact 6.1). Because its linearization is exponentially stable, by Theorem 8.3 $z_2^* = \begin{bmatrix} \pi & 0 \end{bmatrix}^T$ is also exponentially stable for $\dot{z} = \omega(0, z)$.

¹ Though there are technically infinite equilibrium points, due to wrapping, equilibria will be equivalent either of the two presented equilibrium points.

1.3 Output dynamics

The controller proposed by Theorem 20.1 will take the form

$$u = \frac{1}{L_g L_f h(\Phi^{-1}(\eta, z))} (-L_f^2 h(\Phi^{-1}(\eta, z)) - \alpha_2 L_f h(\Phi^{-1}(\eta, z)) - \alpha_1 h(\Phi^{-1}(\eta, z)))$$

Our closed loop output dynamics with this controller becomes

$$\dot{\eta} = \begin{bmatrix} 0 & 1 \\ -\alpha_1 & -\alpha_2 \end{bmatrix} \eta$$

with characteristic equation

$$\lambda^2 + \alpha_2 \lambda + \alpha_1 = 0$$

which has solutions with negative real parts when $\alpha_1, \alpha_2 \in \mathbb{R}_{>0}$. Choosing such α results in $\eta^* = 0$ being an exponentially stable equilibrium of $\dot{\eta} = A_{cl}\eta$.

1.4 Controller simulation

By Theorem 20.1, the system $\dot{x} = f(x) + g(x)k(x)$ is exponentially stable about the equilibrium point $x^* = \Phi^{-1}(\eta^*, z^*)$ with controller

$$k(x) = \frac{1}{L_g L_f h(x)} (-L_f^2 h(x) - \alpha_2 L_f h(x) - \alpha_1 h(x))$$

We can implement this controller in MATLAB and simulate the closed-loop system dynamics. One such realization is shown in Figure 1.

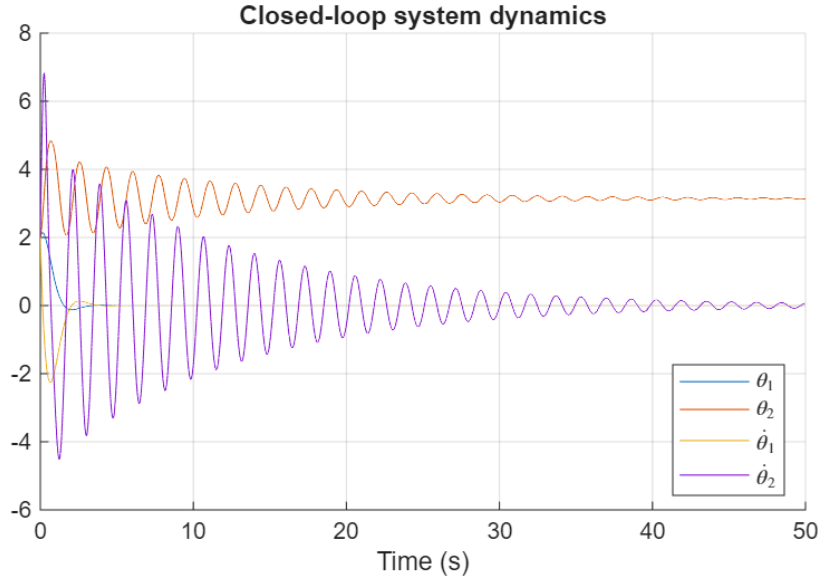


Figure 1: Closed-loop system dynamics with $x_0 = \begin{bmatrix} 2 & 2 & 2 & 2 \end{bmatrix}^T$ and $\begin{bmatrix} \alpha_1 & \alpha_2 \end{bmatrix} = \begin{bmatrix} 5 & 3 \end{bmatrix}$.

We can observe that each state exponentially converges to $x^* = \Phi^{-1}(\eta^*, z^*)$. We are interested in the output and zero dynamics of our system. We can compare the behavior between changes in initial conditions and changes in control gains α_i .

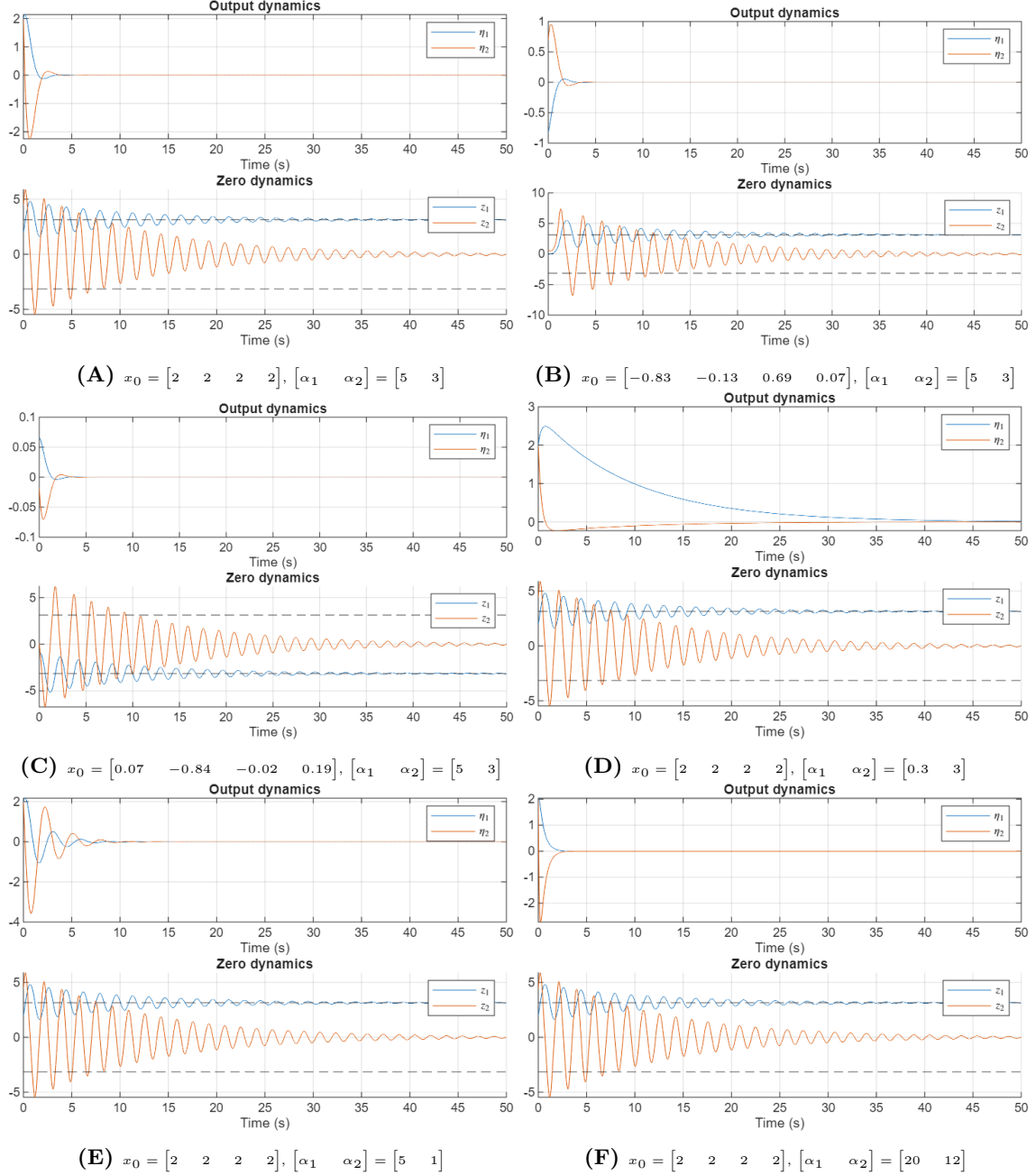


Figure 2: Normal form dynamics with varying initial conditions and control gains. Dashed black lines indicate π and $-\pi$, which are equivalent equilibrium values for $z_1 = \theta_2$.

We can see certain trends from Figure 2. We can observe that the zero dynamics are independent of the control gains — simulations with the same initial condition exhibit the same zero dynamics. The zero dynamics only demonstrate the free behavior of the second pendulum of the system, whose dynamics are only determined by its initial conditions. In addition, the control gains act like a tunable PD controller for the output dynamics, with α_1 dictating the rate of convergence for $\eta_1 = \theta_1$ and α_2 doing so for $\eta_2 = \theta_1$.

2 Appendix

2.1 System

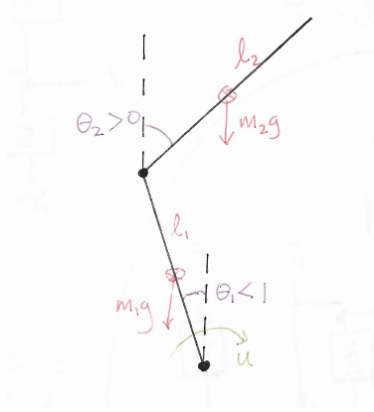


Figure 3: Double inverted pendulum.

An input torque u is applied to the first link of the pendulum. Each arm is modeled as a uniform rod ($I_i = \frac{1}{12}m_i l_i^2$) and friction between the rods is assumed as $|F_1| = b_1|\dot{\theta}_1|$ and $|F_2| = b_2|\dot{\theta}_1 - \dot{\theta}_2|$. Using the angular deviation from the vertical and angular velocities of the links, we can define the state $x = [\theta_1 \ \theta_2 \ \dot{\theta}_1 \ \dot{\theta}_2]^\top$. The dynamics of the system are driven by the following differential equation in the form $\dot{x} = f(x, u)$:

$$f(x, u) = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & \mu_2(x) & \mu_3(x) \\ 0 & 0 & \mu_3(x) & \mu_1(x) \end{bmatrix} \left(\begin{bmatrix} \nu_1(x) \\ \nu_2(x) \\ \nu_3(x) \\ \nu_4(x) \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ 1 \\ 0 \end{bmatrix} u \right) = \begin{bmatrix} \nu_1(x) \\ \nu_2(x) \\ \mu_2(x)\nu_3(x) + \mu_3(x)\nu_4(x) \\ \mu_3(x)\nu_3(x) + \mu_1(x)\nu_4(x) \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ \mu_2(x) \\ \mu_3(x) \end{bmatrix} u$$

2.2 System equations

$$\begin{aligned} \mu_2(x) &= \frac{M_2}{|M(x)|}, \quad \mu_3(x) = \frac{-M_3 \cos \Delta}{|M(x)|}, \quad \mu_1(x) = \frac{M_1}{|M(x)|} \\ \nu_1(x) &= \dot{\theta}_1, \quad \nu_2(x) = \dot{\theta}_2 \\ \nu_3(x) &= -K_1 \dot{\theta}_2^2 \sin \Delta + K_2 \sin \theta_1 - B_1 \dot{\theta}_1 + B_2 \dot{\theta}_2 \\ \nu_4(x) &= K_3 \dot{\theta}_1^2 \sin \Delta + K_4 \sin \theta_2 + B_2 \dot{\theta}_1 - B_2 \dot{\theta}_2 \end{aligned}$$

with parameters defined in Table 1.

M_1	$(\frac{1}{3}m_1 + m_2) l_1^2$
M_2	$\frac{1}{3}m_2 l_2^2$
M_3	$\frac{1}{2}m_2 l_1 l_2$
K_1	$\frac{1}{2}m_2 l_1 l_2$
K_2	$(\frac{1}{2}m_1 + m_2) g l_1$
K_3	$\frac{1}{2}m_2 l_1 l_2$
K_4	$\frac{1}{2}m_2 g l_2$
B_1	$(b_1 + b_2)$
B_2	b_2
Δ	$\theta_1 - \theta_2$
$ M(x) $	$M_1 M_2 - M_3^2 \cos^2 \Delta$

Table 1: Coefficients for rest of derivations.

2.3 Jacobian

The Jacobian of the system is:

$$Df(x) = \begin{bmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ Df_{31}(x) & Df_{32}(x) & Df_{33}(x) & Df_{34}(x) \\ Df_{41}(x) & Df_{42}(x) & Df_{43}(x) & Df_{44}(x) \end{bmatrix}$$

where the elements are defined by the equations below:

$$\begin{aligned} Df_{31}(x) &= \frac{\partial \mu_2(x)}{\partial \theta_1} \nu_3(x) + \mu_2(x) \frac{\partial \nu_3(x)}{\partial \theta_1} + \frac{\partial \mu_3(x)}{\partial \theta_1} \nu_4(x) + \mu_3(x) \frac{\partial \nu_4(x)}{\partial \theta_1} \\ Df_{32}(x) &= \frac{\partial \mu_2(x)}{\partial \theta_2} \nu_3(x) + \mu_2(x) \frac{\partial \nu_3(x)}{\partial \theta_2} + \frac{\partial \mu_3(x)}{\partial \theta_2} \nu_4(x) + \mu_3(x) \frac{\partial \nu_4(x)}{\partial \theta_2} \\ Df_{33}(x) &= \mu_2(x) \frac{\partial \nu_3(x)}{\partial \dot{\theta}_1} + \mu_3(x) \frac{\partial \nu_4(x)}{\partial \dot{\theta}_1} \\ Df_{34}(x) &= \mu_2(x) \frac{\partial \nu_3(x)}{\partial \dot{\theta}_2} + \mu_3(x) \frac{\partial \nu_4(x)}{\partial \dot{\theta}_2} \\ Df_{41}(x) &= \frac{\partial \mu_3(x)}{\partial \theta_1} \nu_3(x) + \mu_3(x) \frac{\partial \nu_3(x)}{\partial \theta_1} + \frac{\partial \mu_1(x)}{\partial \theta_1} \nu_4(x) + \mu_1(x) \frac{\partial \nu_4(x)}{\partial \theta_1} \\ Df_{42}(x) &= \frac{\partial \mu_3(x)}{\partial \theta_2} \nu_3(x) + \mu_3(x) \frac{\partial \nu_3(x)}{\partial \theta_2} + \frac{\partial \mu_1(x)}{\partial \theta_2} \nu_4(x) + \mu_1(x) \frac{\partial \nu_4(x)}{\partial \theta_2} \\ Df_{43}(x) &= \mu_3(x) \frac{\partial \nu_3(x)}{\partial \dot{\theta}_1} + \mu_1(x) \frac{\partial \nu_4(x)}{\partial \dot{\theta}_1} \\ Df_{44}(x) &= \mu_3(x) \frac{\partial \nu_3(x)}{\partial \dot{\theta}_2} + \mu_1(x) \frac{\partial \nu_4(x)}{\partial \dot{\theta}_2} \end{aligned}$$

$$D \begin{bmatrix} \mu_1(x) \\ \mu_2(x) \\ \mu_3(x) \end{bmatrix} = \begin{bmatrix} -M_1 M_3 \frac{\cos \Delta \sin \Delta}{|M(x)|^2} & M_1 M_3 \frac{\cos \Delta \sin \Delta}{|M(x)|^2} & 0 & 0 \\ -M_2 M_3 \frac{\cos \Delta \sin \Delta}{|M(x)|^2} & M_2 M_3 \frac{\cos \Delta \sin \Delta}{|M(x)|^2} & 0 & 0 \\ -M_3 \frac{(2M_3 \cos^2 \Delta - |M(x)|) \sin \Delta}{|M(x)|^2} & M_3 \frac{(2M_3 \cos^2 \Delta - |M(x)|) \sin \Delta}{|M(x)|^2} & 0 & 0 \end{bmatrix}$$

$$D \begin{bmatrix} \nu_1(x) \\ \nu_2(x) \\ \nu_3(x) \\ \nu_4(x) \end{bmatrix} = \begin{bmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ -K_1 \dot{\theta}_2^2 \cos \Delta + K_2 \cos \theta_1 & K_1 \dot{\theta}_2^2 \cos \Delta & -B_1 & -2K_1 \dot{\theta}_2 \sin \Delta + B_2 \\ K_3 \dot{\theta}_1^2 \cos \Delta & -K_3 \dot{\theta}_1^2 \cos \Delta + K_4 \cos \theta_2 & 2K_3 \dot{\theta}_1 \sin \Delta + B_2 & -B_2 \end{bmatrix}$$

Homework 5

The goal of this homework is to use a Control Lyapunov Function and a Quadratic Program to design a stabilizing controller for our system. A process that we can follow is outlined in Remark 24.2. The process is based on Theorem 24.1, which provides a nice recipe for a stabilizing controller for a system, requiring:

- a nonlinear system in normal form
- $z^* = 0$ an exponentially stable equilibrium of the zero dynamics $\dot{z} = \omega(0, z)$
- existence of a CLF on the output dynamics, $V : Y \rightarrow \mathbb{R}$
- any locally Lipschitz continuous feedback control law $u = k(\eta, z) \in K(\eta, z)$, where $K(\eta, z)$ is the set of control values:

$$K(\eta, z) \triangleq \left\{ u \in U : L_{\hat{f}}V(\eta, z) + L_{\hat{g}}V(\eta, z)u + \lambda_3 V(\eta) \leq 0 \right\}$$

The fulfillment of these requirements renders (η^*, z^*) an exponentially stable equilibrium point for the following closed-loop system:

$$\begin{bmatrix} \dot{\eta} \\ \dot{z} \end{bmatrix} = \hat{f}_{\text{cl}} \triangleq \begin{bmatrix} \hat{f}(\eta, z) + \hat{g}(\eta, z)k(\eta, z) \\ \omega(\eta, z) \end{bmatrix}$$

It follows that we can turn the system $\dot{x} = f(x) + g(x)u$, with $x \in E \subseteq \mathbb{R}^n$ and $u \in U \subseteq \mathbb{R}^m$, can be transformed into normal form using the diffeomorphism $\Phi : E \rightarrow Y \times Z$. We meet the requirements for Theorem 24.1 and then find a suitable controller that operates on $(\eta, z) = \Phi(x)$, which stabilizes the system in normal form. Because the zero dynamics are exponentially stable, $x^* = \Phi^{-1}(0, 0)$ in the original form of the dynamics is exponentially stable (cf. Claim 20.1).

In the previous homework assignment, we derived the first two items for a SISO system with output $y = h(x) = \theta_1$, which drives the first link to an upright position. Thus, the remainder of this assignment will be to find a CLF for the output dynamics and to find a locally Lipschitz continuous feedback control law.

1 Control Lyapunov Function

From the previous assignment, through feedback linearization we had obtained a controller that rendered our closed-loop output dynamics into

$$\dot{\eta} = \begin{bmatrix} 0 & 1 \\ \alpha_1 & \alpha_2 \end{bmatrix} \eta = A_{\text{cl}} \eta$$

where choosing $\alpha_1, \alpha_2 \in \mathbb{R}_{>0}$ renders the origin exponentially stable. Provided any positive definite matrix Q , the solution $P = P^T > 0$ to the CTLE

$$A_{\text{cl}}^T P + P A_{\text{cl}} = -Q$$

can be used to construct a Lyapunov function $V(\eta) = \eta^T P \eta$ that certifies the exponential stability of this system. By Lemma 24.1, this Lyapunov function is also a control Lyapunov function. Thus, for any (locally) Lipschitz continuous feedback control law $u = k(\eta, z) \in K(\eta, z)$ renders the equilibrium point $(\eta^*, z^*) = (0, 0)$ exponentially stable for our closed-loop dynamics in normal form.

2 Quadratic Program control law

Now we just need a locally Lipschitz continuous feedback control law. From feedback linearization, we have a control law in the form of $u = \mathcal{G}^{-1}(\Phi^{-1}(\eta, z))(-\mathcal{F}(\Phi^{-1}(\eta, z)) + v)$. Lecture 24 proves that the following quadratic program (v -CLF-QP) is locally Lipschitz continuous on Y :

$$\begin{aligned} v^*(\eta, z) = \operatorname{argmin}_{v \in \mathbb{R}^m} v^T v \\ \text{s.t. } \eta^T (F^T P + P F) \eta + 2\eta^T P G v \leq -\frac{\lambda_{\min}(Q)}{\lambda_{\max}(P)} \eta^T P \eta \end{aligned}$$

Because $\mathcal{G}^{-1}$ and $\mathcal{F}$ are also locally Lipschitz continuous,

$$u = k^*(\eta, z) = \mathcal{G}^{-1}(\Phi^{-1}(\eta, z))(-\mathcal{F}(\Phi^{-1}(\eta, z)) + v^*(\eta, z)) \quad (1)$$

is also locally Lipschitz continuous function on Y . Thus by Theorem 24.1, applying this controller¹ will establish an exponentially stable equilibrium point for the closed-loop system dynamics.

Now that we know that this controller will stabilize our system, we can work around with v -CLF-QP to a more suitable form for implementation. Starting with the minimizing argument:

$$\begin{aligned} \eta^T (F^T P + P F) \eta + 2\eta^T P G v \leq -\frac{\lambda_{\min}(Q)}{\lambda_{\max}(P)} \eta^T P \eta \\ \eta^T \left(F^T P + P F + \frac{\lambda_{\min}(Q)}{\lambda_{\max}(P)} P \right) \eta + 2\eta^T P G v \leq 0 \end{aligned}$$

We can use this to formulate the solution $v^*(\eta, z)$ in closed form:

$$v^*(\eta, z) = \begin{cases} -\frac{\eta^T \left(F^T P + P F + \frac{\lambda_{\min}(Q)}{\lambda_{\max}(P)} P \right) \eta}{2\eta^T P G G^T P \eta} G^T P \eta & \text{if } \eta^T \left(F^T P + P F + \frac{\lambda_{\min}(Q)}{\lambda_{\max}(P)} P \right) \eta > 0 \\ 0 & \text{if } \eta^T \left(F^T P + P F + \frac{\lambda_{\min}(Q)}{\lambda_{\max}(P)} P \right) \eta \leq 0 \end{cases}$$

Once we obtain $v^*(\eta, z)$, we can use Equation 1 to obtain our system's control input.

3 Simulation

Approach

For a typical quadratic program in the form

$$u = k(x) = \operatorname{argmin}_{u \in \mathbb{R}^m} u^T H(x) u$$

the matrix $H(x)$ can be used to alter the weights of the control effort and influence the behavior of the closed-loop as it converges to its equilibrium point. However, because we are working with a SISO system, the minimizing argument does not change with a choice in H . Thus, we will instead work with changing $Q = Q^T > 0$ when solving the CTLE

$$A_{\text{cl}}^T P + P A_{\text{cl}} = -Q$$

¹ It is necessary to assume $U = \mathbb{R}^m$. We will apply this assumption for our design and simulation.

This will result in different solutions $P = P^T > 0$ and alter the minimizing argument for the v -CLF-QP. To focus our results on changes in Q , we should fix A_{cl} for all simulations. This choice is somewhat arbitrary, so long as A_{cl} is Hurwitz, so we will pick one from the previous homework.

$$A_{cl} = \begin{bmatrix} 0 & 1 \\ -5 & -3 \end{bmatrix}$$

For our simulations, we will work with the original system $\dot{x} = f(x) + g(x)u$. We will then apply the diffeomorphism $\Phi : E \rightarrow Y \times Z$ such that $(\eta, z) = \Phi(x)$. This new coordinate will then be fed into $u = k^*(\Phi(x)) = \mathcal{G}^{-1}(x)(-\mathcal{F}(x) + v^*(\Phi(x)))$, which will be used for the closed-loop system $\dot{x} = f(x) + g(x)k^*(\Phi(x))$.

3.1 Results

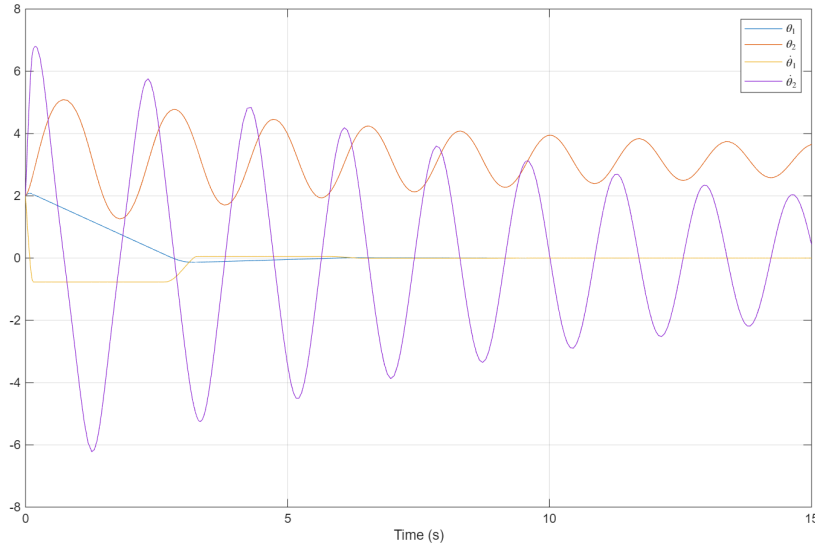


Figure 1: Closed-system behavior with $x_0 = \begin{bmatrix} 2 & 2 & 2 & 2 \end{bmatrix}^T$ and $Q = I$.

The first simulation in Figure 1 demonstrate the closed-loop system with $Q = I$ starting from an arbitrary initial state. We see that the output states η (θ_1 in blue and $\dot{\theta}_1$ in yellow) are quickly driven to zero, driving the output dynamics to its equilibrium point. The normal dynamics thus behave like zero dynamics, and slowly decays to its exponentially stable equilibrium point. This demonstrates that our controller drives our system into asymptotic stability, as predicted by our analysis. Now we can experiment with varying Q for the CTLE. Note: with the assumption of exponentially stable zero dynamics, we will not show them in our simulation result as to focus on the performance of our controller.

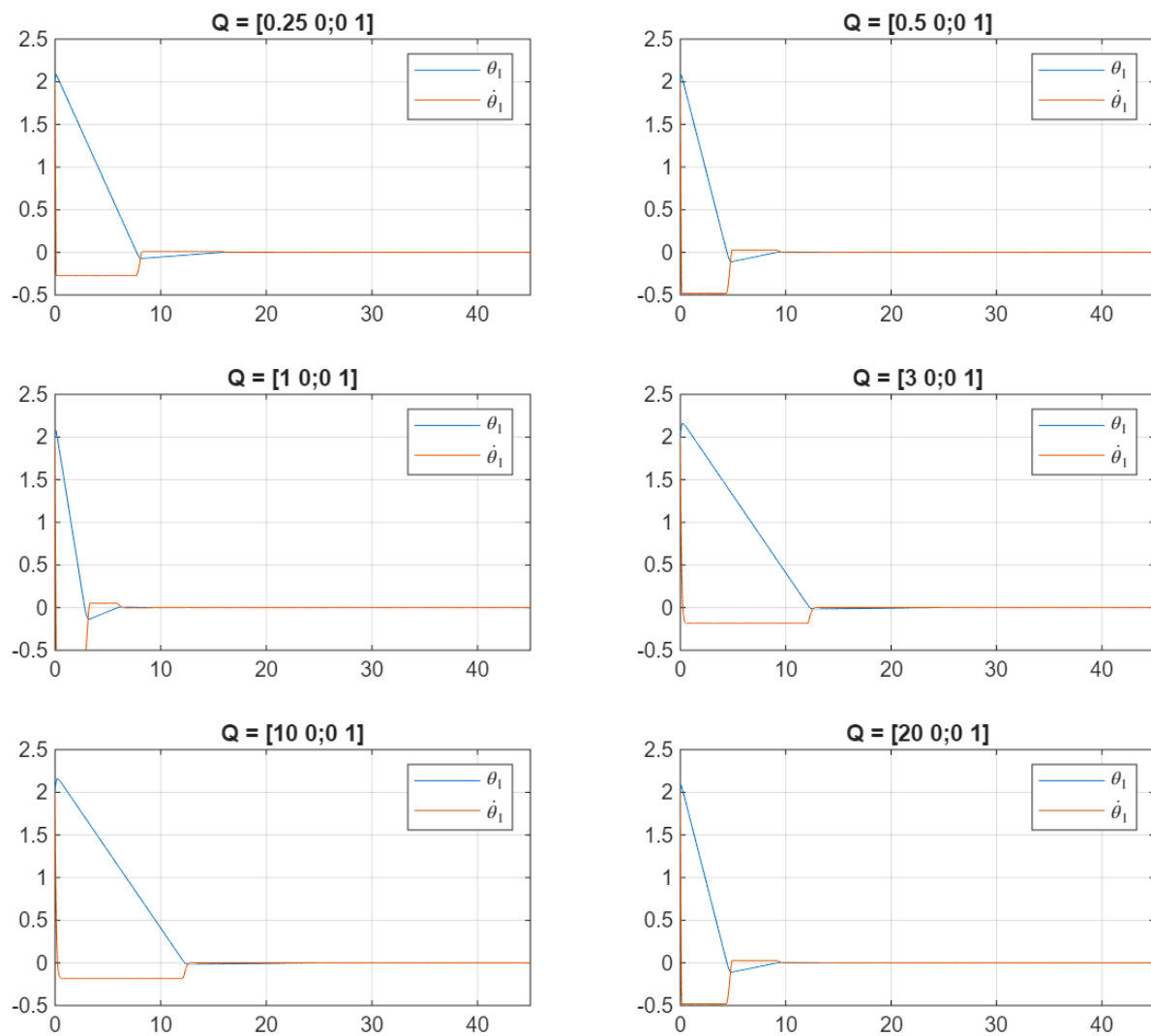


Figure 2: Closed-loop behavior with scaling (1,1) value of Q with $x_0 = \begin{bmatrix} 2 & 2 & 2 & 2 \end{bmatrix}^T$.

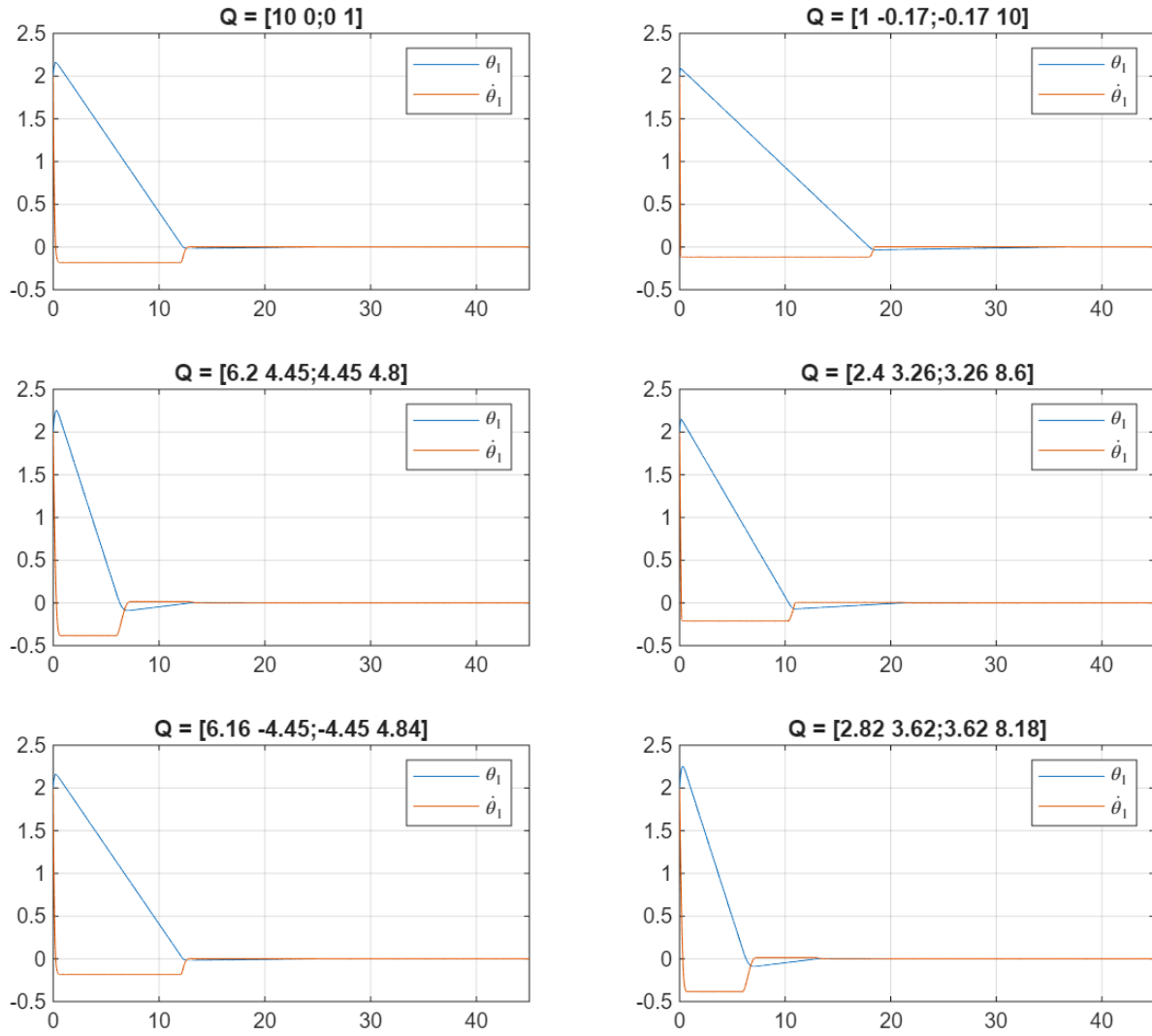


Figure 3: Closed-loop behavior varying Q with same eigenvalues, with $x_0 = \begin{bmatrix} 2 & 2 & 2 & 2 \end{bmatrix}^T$.

Figures 2 and 3 demonstrate the closed-loop behavior with different input matrices to the CTLE. We see that different choices of Q alter the convergence rate of the system. If we look at the v -CLF,

$$L_F V(x) + L_G V(x)v \leq -\frac{\lambda_{\min}(Q)}{\lambda_{\max}(P)} V(x)$$

the choice of Q and the resulting P directly influence the bounds of $\dot{V}(x)$. Regardless of the choice, we can see that the system always converges to the equilibrium point, demonstrating that asymptotic stability is not destroyed.

4 Appendix

4.1 System

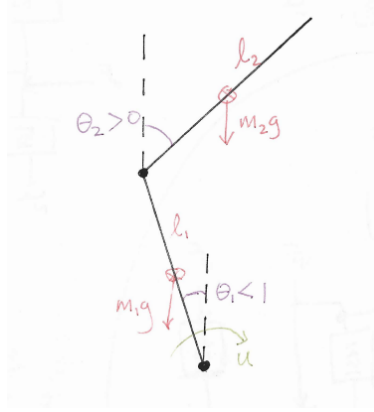


Figure 4: Double inverted pendulum.

An input torque u is applied to the first link of the pendulum. Each arm is modeled as a uniform rod ($I_i = \frac{1}{12}m_i l_i^2$) and friction between the rods is assumed as $|F_1| = b_1|\dot{\theta}_1|$ and $|F_2| = b_2|\dot{\theta}_1 - \dot{\theta}_2|$. Using the angular deviation from the vertical and angular velocities of the links, we can define the state $x = [\theta_1 \ \theta_2 \ \dot{\theta}_1 \ \dot{\theta}_2]^\top$. The dynamics of the system are driven by the following differential equation in the form $\dot{x} = f(x, u)$:

$$f(x, u) = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & \mu_2(x) & \mu_3(x) \\ 0 & 0 & \mu_3(x) & \mu_1(x) \end{bmatrix} \left(\begin{bmatrix} \nu_1(x) \\ \nu_2(x) \\ \nu_3(x) \\ \nu_4(x) \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ 1 \\ 0 \end{bmatrix} u \right) = \begin{bmatrix} \nu_1(x) \\ \nu_2(x) \\ \mu_2(x)\nu_3(x) + \mu_3(x)\nu_4(x) \\ \mu_3(x)\nu_3(x) + \mu_1(x)\nu_4(x) \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ \mu_2(x) \\ \mu_3(x) \end{bmatrix} u$$

4.2 System equations

$$\begin{aligned} \mu_2(x) &= \frac{M_2}{|M(x)|}, \quad \mu_3(x) = \frac{-M_3 \cos \Delta}{|M(x)|}, \quad \mu_1(x) = \frac{M_1}{|M(x)|} \\ \nu_1(x) &= \dot{\theta}_1, \quad \nu_2(x) = \dot{\theta}_2 \\ \nu_3(x) &= -K_1 \dot{\theta}_2^2 \sin \Delta + K_2 \sin \theta_1 - B_1 \dot{\theta}_1 + B_2 \dot{\theta}_2 \\ \nu_4(x) &= K_3 \dot{\theta}_1^2 \sin \Delta + K_4 \sin \theta_2 + B_2 \dot{\theta}_1 - B_2 \dot{\theta}_2 \end{aligned}$$

with parameters defined in Table 1.

M_1	$(\frac{1}{3}m_1 + m_2) l_1^2$
M_2	$\frac{1}{3}m_2 l_2^2$
M_3	$\frac{1}{2}m_2 l_1 l_2$
K_1	$\frac{1}{2}m_2 l_1 l_2$
K_2	$(\frac{1}{2}m_1 + m_2) g l_1$
K_3	$\frac{1}{2}m_2 l_1 l_2$
K_4	$\frac{1}{2}m_2 g l_2$
B_1	$(b_1 + b_2)$
B_2	b_2
Δ	$\theta_1 - \theta_2$
$ M(x) $	$M_1 M_2 - M_3^2 \cos^2 \Delta$

Table 1: Coefficients for rest of derivations.

4.3 Jacobian

The Jacobian of the system is:

$$Df(x) = \begin{bmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ Df_{31}(x) & Df_{32}(x) & Df_{33}(x) & Df_{34}(x) \\ Df_{41}(x) & Df_{42}(x) & Df_{43}(x) & Df_{44}(x) \end{bmatrix}$$

where the elements are defined by the equations below:

$$\begin{aligned} Df_{31}(x) &= \frac{\partial \mu_2(x)}{\partial \theta_1} \nu_3(x) + \mu_2(x) \frac{\partial \nu_3(x)}{\partial \theta_1} + \frac{\partial \mu_3(x)}{\partial \theta_1} \nu_4(x) + \mu_3(x) \frac{\partial \nu_4(x)}{\partial \theta_1} \\ Df_{32}(x) &= \frac{\partial \mu_2(x)}{\partial \theta_2} \nu_3(x) + \mu_2(x) \frac{\partial \nu_3(x)}{\partial \theta_2} + \frac{\partial \mu_3(x)}{\partial \theta_2} \nu_4(x) + \mu_3(x) \frac{\partial \nu_4(x)}{\partial \theta_2} \\ Df_{33}(x) &= \mu_2(x) \frac{\partial \nu_3(x)}{\partial \dot{\theta}_1} + \mu_3(x) \frac{\partial \nu_4(x)}{\partial \dot{\theta}_1} \\ Df_{34}(x) &= \mu_2(x) \frac{\partial \nu_3(x)}{\partial \dot{\theta}_2} + \mu_3(x) \frac{\partial \nu_4(x)}{\partial \dot{\theta}_2} \\ Df_{41}(x) &= \frac{\partial \mu_3(x)}{\partial \theta_1} \nu_3(x) + \mu_3(x) \frac{\partial \nu_3(x)}{\partial \theta_1} + \frac{\partial \mu_1(x)}{\partial \theta_1} \nu_4(x) + \mu_1(x) \frac{\partial \nu_4(x)}{\partial \theta_1} \\ Df_{42}(x) &= \frac{\partial \mu_3(x)}{\partial \theta_2} \nu_3(x) + \mu_3(x) \frac{\partial \nu_3(x)}{\partial \theta_2} + \frac{\partial \mu_1(x)}{\partial \theta_2} \nu_4(x) + \mu_1(x) \frac{\partial \nu_4(x)}{\partial \theta_2} \\ Df_{43}(x) &= \mu_3(x) \frac{\partial \nu_3(x)}{\partial \dot{\theta}_1} + \mu_1(x) \frac{\partial \nu_4(x)}{\partial \dot{\theta}_1} \\ Df_{44}(x) &= \mu_3(x) \frac{\partial \nu_3(x)}{\partial \dot{\theta}_2} + \mu_1(x) \frac{\partial \nu_4(x)}{\partial \dot{\theta}_2} \end{aligned}$$

$$D \begin{bmatrix} \mu_1(x) \\ \mu_2(x) \\ \mu_3(x) \end{bmatrix} = \begin{bmatrix} -M_1 M_3 \frac{\cos \Delta \sin \Delta}{|M(x)|^2} & M_1 M_3 \frac{\cos \Delta \sin \Delta}{|M(x)|^2} & 0 & 0 \\ -M_2 M_3 \frac{\cos \Delta \sin \Delta}{|M(x)|^2} & M_2 M_3 \frac{\cos \Delta \sin \Delta}{|M(x)|^2} & 0 & 0 \\ -M_3 \frac{(2M_3 \cos^2 \Delta - |M(x)|) \sin \Delta}{|M(x)|^2} & M_3 \frac{(2M_3 \cos^2 \Delta - |M(x)|) \sin \Delta}{|M(x)|^2} & 0 & 0 \end{bmatrix}$$

$$D \begin{bmatrix} \nu_1(x) \\ \nu_2(x) \\ \nu_3(x) \\ \nu_4(x) \end{bmatrix} = \begin{bmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ -K_1 \dot{\theta}_2^2 \cos \Delta + K_2 \cos \theta_1 & K_1 \dot{\theta}_2^2 \cos \Delta & -B_1 & -2K_1 \dot{\theta}_2 \sin \Delta + B_2 \\ K_3 \dot{\theta}_1^2 \cos \Delta & -K_3 \dot{\theta}_1^2 \cos \Delta + K_4 \cos \theta_2 & 2K_3 \dot{\theta}_1 \sin \Delta + B_2 & -B_2 \end{bmatrix}$$

Homework 6

1 ISS-CLF controller

We want to construct an ISS-CLF and implement the corresponding controller via a quadratic program. We can use our control Lyapunov function from last time and a u -CLF-QP controller $u = k(x)$. These two together verify that

$$\dot{V}(x) = L_f V(x) + L_g V(x) \leq -\alpha(\|x\|)$$

for the system $\dot{x} = f(x) + g(x)k(x)$. Remark 27.2 argues that this Lyapunov function is also an ISS-CLF for the disturbed system $\dot{x} = f(x) + g(x)u + g(x)d$ with the altered control law

$$u = k(x) - \frac{1}{\varepsilon} L_g V(x)^T$$

where we define $\varepsilon > 0$ as a design parameter. As performed in Chapter 27.2, we can use the following quadratic program to establish ISS stability for our system:

$$\begin{aligned} k_{\text{ISS}}(x) = \underset{u \in \mathbb{R}^m}{\operatorname{argmin}} \quad & \frac{1}{2} u^T H(x) u + F(x)^T u \\ \text{s.t.} \quad & L_f V(x) + L_g V(x) u \leq -\lambda V(x) - \frac{1}{\varepsilon} \|L_g V(x)\|^2 \end{aligned}$$

We can choose $H(x)$ and $F(x)^T$ from the u -CLF-QP, since this choice ensured that the controller was locally Lipschitz continuous on E . Because we are using a Lyapunov function based on the solution to the CARE or CTLE, we also have $\lambda = \lambda_{\min}(Q)/\lambda_{\max}(P)$. Also, just to be very clear, because we based our controller and Lyapunov function on the output dynamics of our system, we should be expecting, at the very least, stability for our output dynamics only and base our assumption for full-system stability stemming from the exponential stability of our zero dynamics.

1.1 Playing with ε

For our controller, ε is a user defined parameter. We want to focus our simulations on varying disturbances to examine the robustness of our system. Thus, we should really try to use the same value through the rest of our simulations.

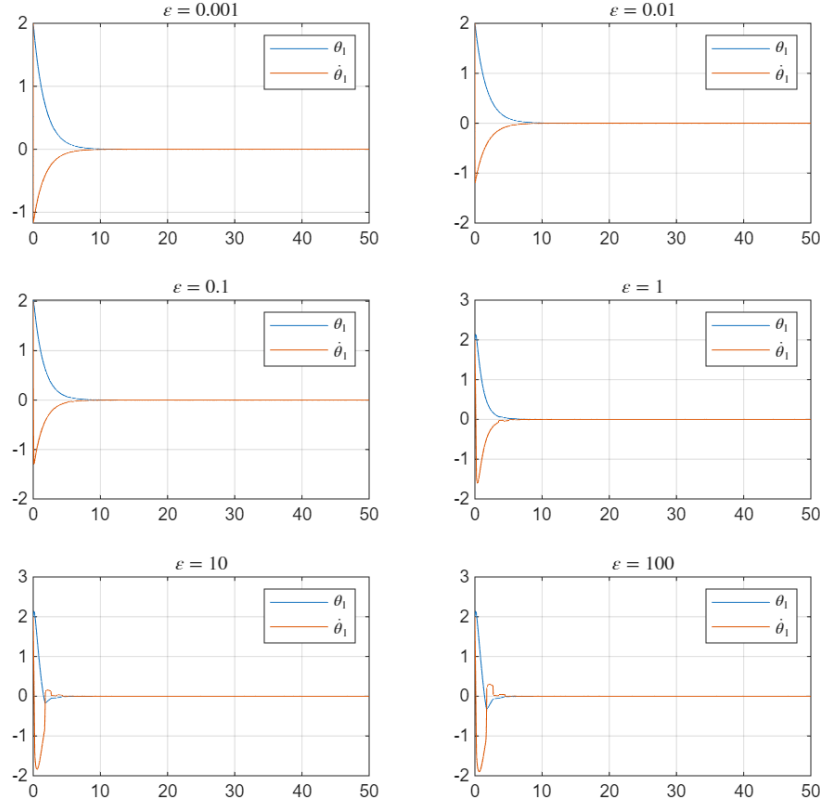


Figure 1: Closed-loop system behavior with varying values of ε with $d(t) = 0$.

The isolated effects of changing ε are shown in Figure 1¹. We see that increasing the value of ε makes the output converge much faster. However, this comes at the price of increased magnitude of the state in its transient response. For the rest of the simulations, we will use $\varepsilon = 0.1$, as it has decent balance convergence rate and magnitude of overshoot.

1.2 Disturbance response and robustness

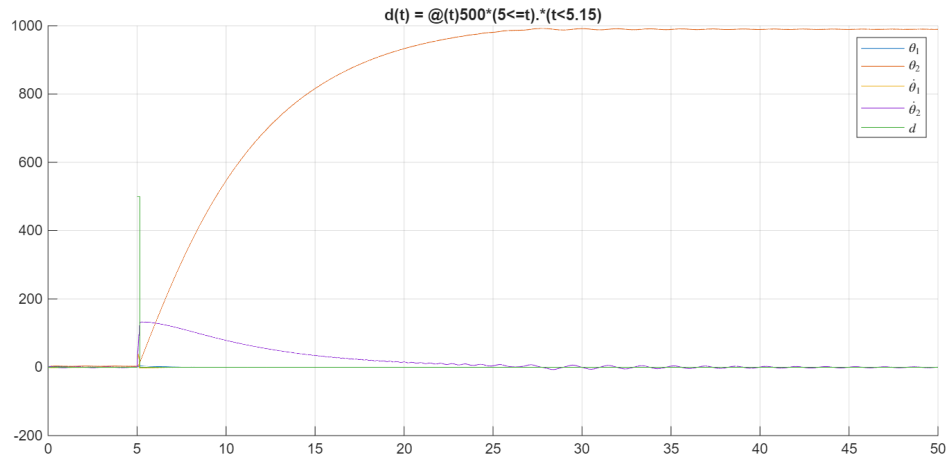


Figure 2: Simulated impulse response from arbitrary initial state.

¹ These results show the behavior of the output dynamics only. This seems fair to do, as we have exponentially stable zero dynamics.

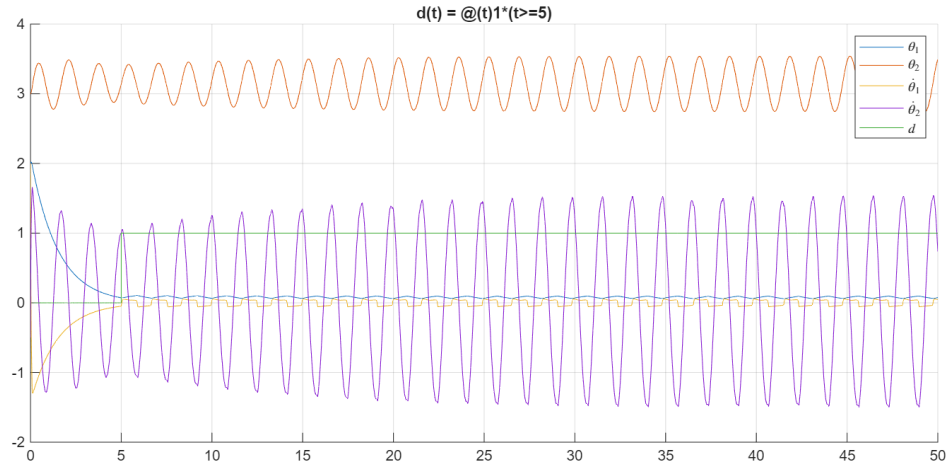


Figure 3: Simulated step response from arbitrary initial state.

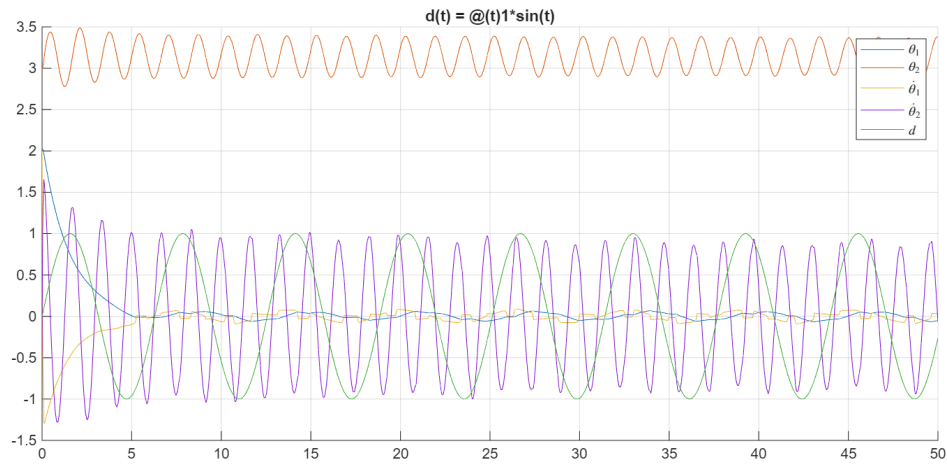


Figure 4: Simulated low frequency response from arbitrary initial state.

The above figures demonstrate the closed-loop system's response to different types of matched disturbances. With an impulse disturbance to the system, Figure 2 shows a large injection of energy to the system before it settles to an equilibrium point of the undisturbed system. However, with persistent disturbance signals, the behavior is a bit different. The output dynamics (blue and yellow lines) in Figures 3 and 4 are driven close, but not exactly, to their equilibrium values. Moreover, it appears that there is some oscillatory behavior to the output dynamics that drive the normal dynamics to also oscillate around their equilibrium values. This is related to what we would expect for ISS system behavior with a disturbance — rather than converging to the equilibrium point, the solution converges to a ball around the equilibrium point instead.

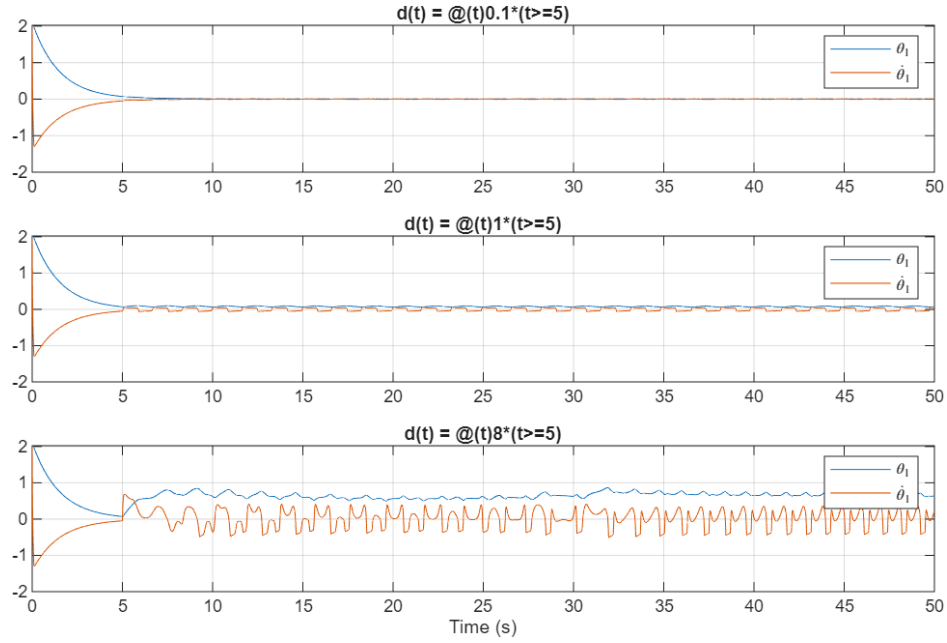


Figure 5: Output dynamics with varying magnitude step disturbance applied at $t = 5$ seconds.

Another idea for ISS systems is that the size of the disturbance directly influences the size of the ball that solutions converge to. This is explored in Figure 5, where we can see that larger magnitude disturbances result in solutions that are farther away from the equilibrium point of the system. We also see some regular oscillations in our output states. This has some consequences when we look at our full-state system.

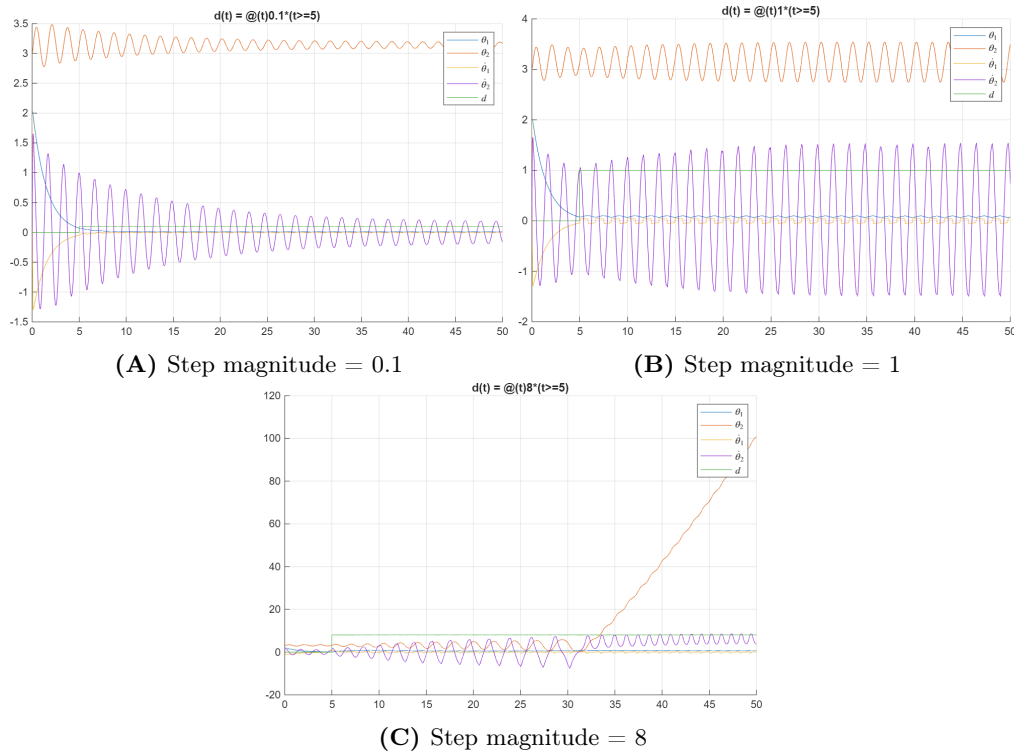


Figure 6: Full-system behavior to varying magnitude step disturbance.

We see in Figure 6 that the behavior of the second link is much different than that of the first link. It seems that the magnitude of the disturbance affects the frequency of oscillation of the first link. It seems that at a certain frequency, there is resonance in the system, and the second link rotates continually, refusing to settle/oscillate about its equilibrium.

2 Safety and CBF

If we want to enforce stability, we need to define a safe set $\mathcal{S}$. We have a Lyapunov function that is valid for on the set of output states Y . We can use this to at least generate a barrier function also defined on Y through Proposition 12.1, where we define a barrier function that establishes the forward invariance of a Lyapunov sublevel set. Thus, we can have the following:

$$\begin{aligned} h(\eta) &= 10 - v(\eta) \\ \mathcal{S} &= \{\eta \in Y \mid h(\eta) \geq 0\} \end{aligned}$$

From here, we can² use the diffeomorphism $\Phi : E \rightarrow Y \times Z$ to get this barrier function and safety set in terms of our original coordinates so that $(\eta, z) = \Phi(x)$.

$$\begin{aligned} h(\Phi(x)) &= 10 - v(\Phi(x)) \\ \mathcal{S} &= \{x \in E \mid h(\Phi(x)) \geq 0\} \end{aligned}$$

We need to check if this is also a valid Control Barrier Function. Because this is based on a quadratic Lyapunov function, it is simple to take the derivative.

$$\begin{aligned} \dot{h}(\eta) &= \frac{d}{dt}(10 - v(\eta)) \\ &= -\dot{v}(\eta) \\ &= -L_F v(\eta) - L_G v(\eta)(\mathcal{G}(\eta)u + \mathcal{F}(\eta)) \\ &\geq \frac{\lambda_{\min}(Q)}{\lambda_{\max}(P)} v(\eta) \\ &\geq -\frac{\lambda_{\min}(Q)}{\lambda_{\max}(P)} \cdot 10 + \frac{\lambda_{\min}(Q)}{\lambda_{\max}(P)} v(\eta) \\ &= -\frac{\lambda_{\min}(Q)}{\lambda_{\max}(P)} (10 - v(\eta)) \\ &= -\frac{\lambda_{\min}(Q)}{\lambda_{\max}(P)} h(\eta) \end{aligned}$$

Thus, $h(\eta)$ is a CBF for our output dynamics. Is this an ISSf-CBF? Remark 27.4 says that it is, so long as we enforce the following condition:

$$\sup_{u \in U} [L_f h(\eta) + L_g h(\eta)u - \frac{1}{\varepsilon} L_g h(\eta) L_g h(\eta)^T] \geq -\alpha(h(\eta))$$

Now that we have both an ISS-CLF and an ISSf-CBF, Chapter 27.3 provides a framework for a ISS-ISSF-QP controller:

² I pray to God that this is something we can do.

$$\begin{aligned}
\mathbf{u}^*(x) = & \underset{\mathbf{u}=(u,\delta) \in \mathbb{R}^{m+1}}{\operatorname{argmin}} \quad \frac{1}{2}u^T H(x)u + F(x)^T u + p\delta \\
\text{s.t.} \quad & L_f \mathbf{V}(x) + L_g \mathbf{V}(x)u \leq -\lambda \mathbf{V}(x) - \frac{1}{\varepsilon_V} \|L_g \mathbf{V}(x)\|^2 + \delta \\
& L_f \mathbf{h}(x) + L_g \mathbf{h}(x)u \geq -\alpha(\mathbf{h}(x)) + \frac{1}{\varepsilon_S} \|L_g \mathbf{h}(x)\|^2
\end{aligned}$$

with $\varepsilon_V, \varepsilon_S > 0$. This controller always ensures safety for the output dynamics, but a relaxation factor δ must be added to ensure that the QP is solvable and continuous. However, if a solution returns where $\delta \approx 0$, it means that both the control objective and barrier function are compatible — asymptotic stability could still be achievable. This is a situation that could be prioritized through increasing p in the cost term, which creates a preference for smaller values of δ . In an attempt to do so, we will set $p = 100$. We will also arbitrarily set $\varepsilon_S = 0.1$.

2.1 Zero-disturbance simulation

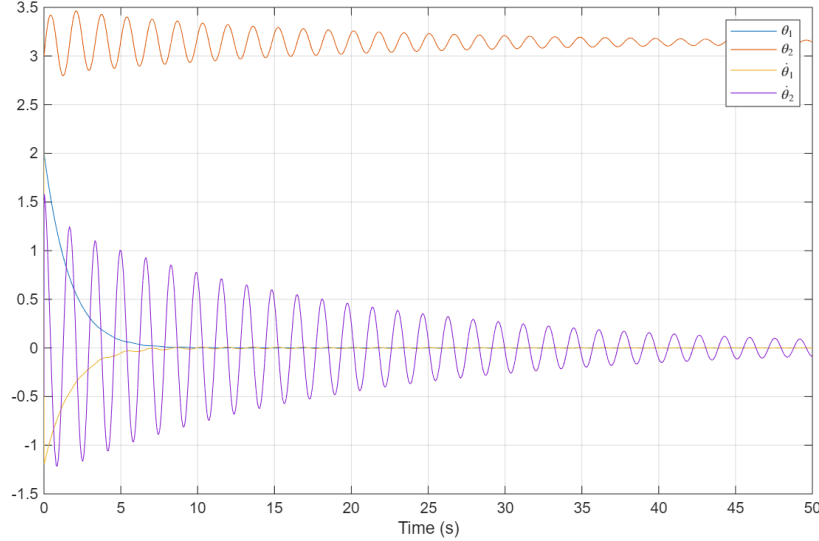


Figure 7: Closed-loop behavior with ISS-ISSf-QP controller starting from arbitrary initial state. No disturbance provided.

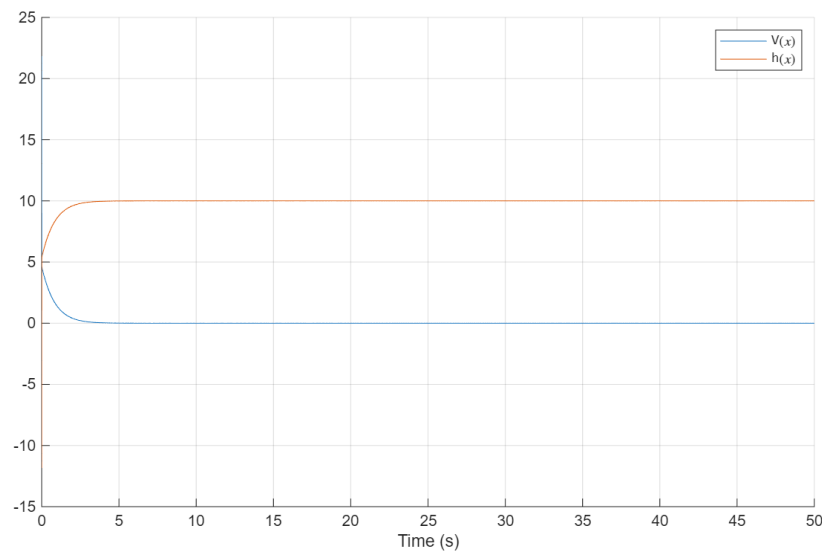


Figure 8: Evolution of $V(x)$ and $h(x)$.

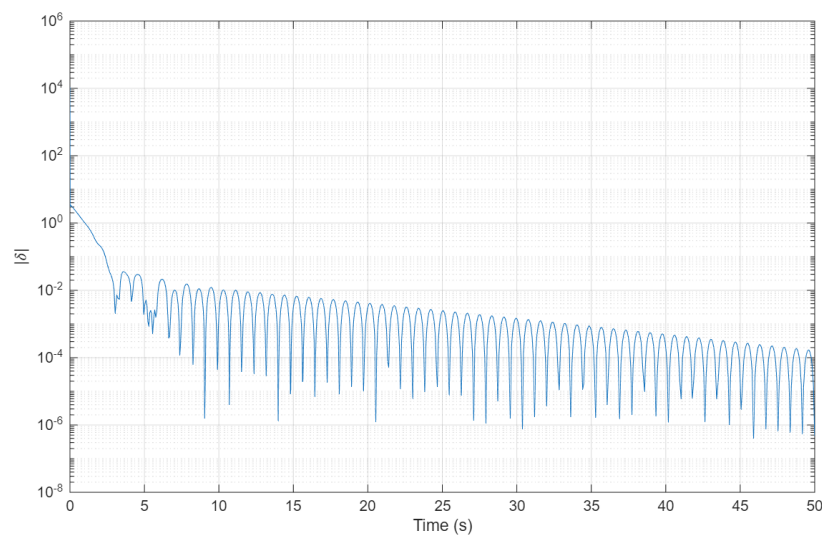


Figure 9: Magnitude of δ from $\mathbf{u}^*(x)$.

These results look nice. We see that the system is quickly driven to the safe set $\mathcal{S}$ via Figure 8, where the values of $h(x)$ is quickly driven to take a value above 0. Moreover, Figures 7-9 show that we preserve asymptotic stability, as the output states and $V(x)$ are driven to zero, as well a small magnitude of δ from the quadratic program.

2.2 Disturbance response

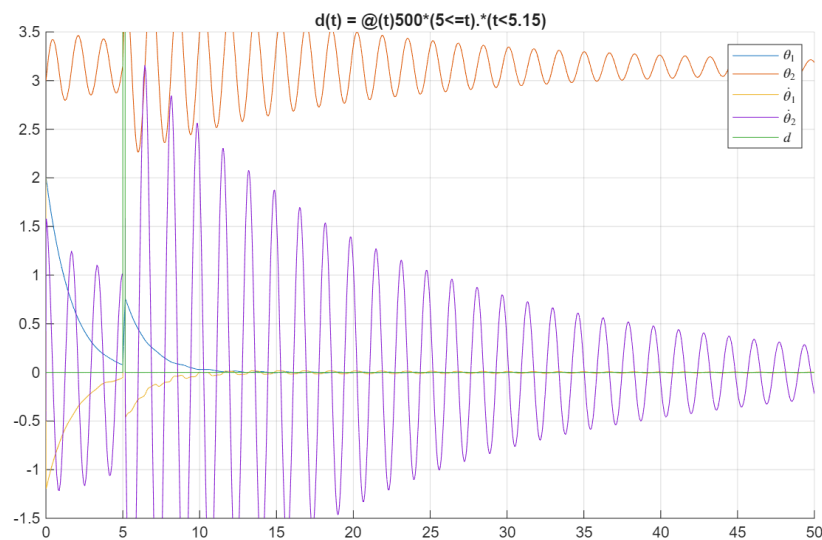


Figure 10: ISS-ISSf-QP controller impulse disturbance behavior.

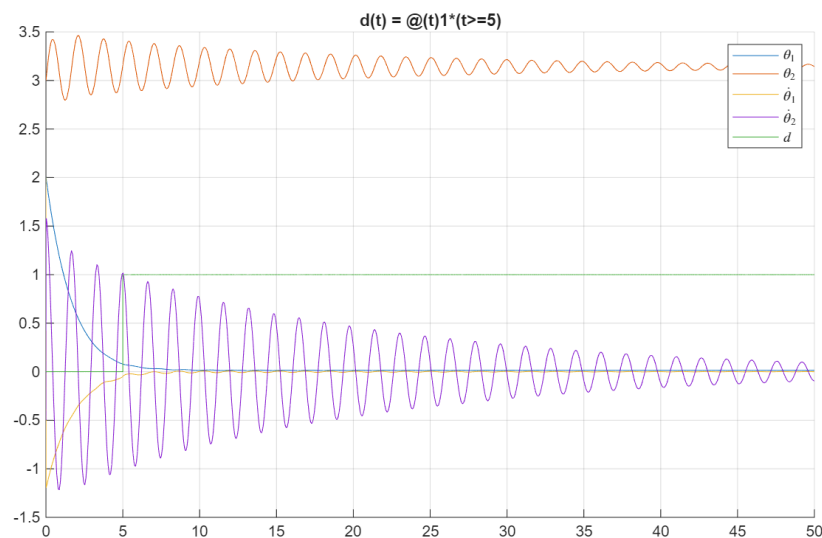


Figure 11: ISS-ISSf-QP controller step disturbance behavior.

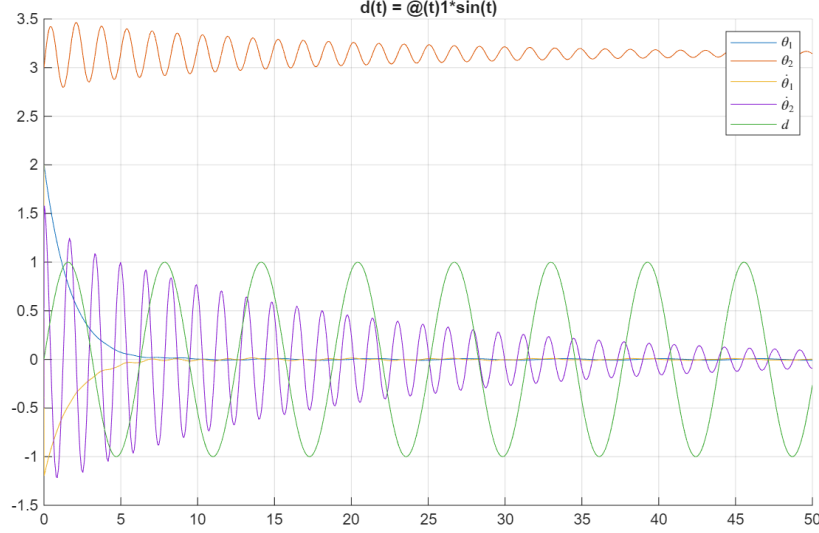
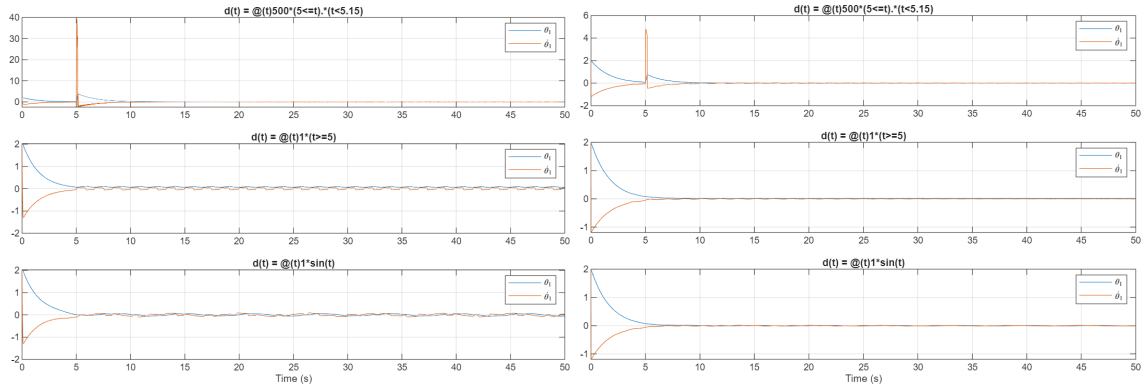


Figure 12: ISS-ISSf-QP controller sinusoidal disturbance behavior.



(A) ISS-QP controller

(B) ISS-ISSf controller

Figure 13: Closed-loop output dynamics with varying disturbance types.

It appears that adding the ISSf-CBF to our quadratic program improved the behavior of our closed-loop system. It seems that the great effort into keeping our output dynamics in a forward invariant set prevents it from reaching troublesome states that have adverse effects on the normal dynamics. For example, Figure 2 shows how an impulse response will cause a large magnitude spike for θ_1 which, in turn, causes the normal dynamic states to be sent flying. However, with safety enforced, 10 prevents this large spike from happening, keeping the normal dynamic states at a low magnitude. We can also see in Figures 11 and 12 that we no longer see odd oscillatory behavior and resonance, possibly due to the relaxation of the ISS-CLF condition. These results are more clearly seen in Figure 13, which isolate the output dynamic states of the two controllers. Thus, it seems that, at least with this specific realization of $h(x)$ and $\mathcal{S}$, we are able to achieve both safety and asymptotic stability in the closed-loop system.

3 Safety and stability conflicts

It is fairly simple to think of a situation, though not very practical, where safety and stability are not compatible. Let's say we keep the ISS-CLF that we have been working

with. However, when we are constructing an ISSf-CBF, let's derive it from a different Lyapunov function — one that establishes exponential equilibrium around a different equilibrium point, say $\eta_{eq} = \begin{bmatrix} 3\pi/4 & 0 \end{bmatrix}^T$ — and use a much smaller sublevel set:

$$\begin{aligned} \mathbf{v}'(\eta) &= (\eta - \eta_{eq})^T P (\eta - \eta_{eq}) \\ \longrightarrow \mathbf{h}'(\eta) &= 0.1 - \mathbf{v}'(\eta) \end{aligned}$$

This new function $\mathbf{h}'(\eta)$ establishes the safety of

$$\mathcal{S}' = \{\eta \in Y \mid \mathbf{h}'(\eta) \geq 0\}$$

through Proposition 12.1. We then use the diffeomorphism Φ to obtain the barrier function and safety set in the original coordinates:

$$\begin{aligned} \mathbf{h}'(\Phi(x)) &= 0.1 - \mathbf{v}'(\Phi(x)) \\ \mathcal{S}' &= \{x \in E \mid \mathbf{h}'(\Phi(x)) \geq 0\} \end{aligned}$$

A similar analysis to before can be performed to demonstrate that this is also an ISSf-CBF, so long as we enforce:

$$\sup_{u \in U} [L_f \mathbf{h}'(\eta) + L_g \mathbf{h}'(\eta)u - \frac{1}{\varepsilon} L_g \mathbf{h}'(\eta) L_g \mathbf{h}'(\eta)^T] \geq -\alpha(\mathbf{h}'(\eta))$$

Now we can try to apply the ISS-ISSf-QP controller.

3.1 Simulation

MATLAB had a hard time with this one. It appears that before the one-second mark, the `quadprog()` function I was using could not find a solution, likely due to something trying to go to infinity. I will present up to the failure point.

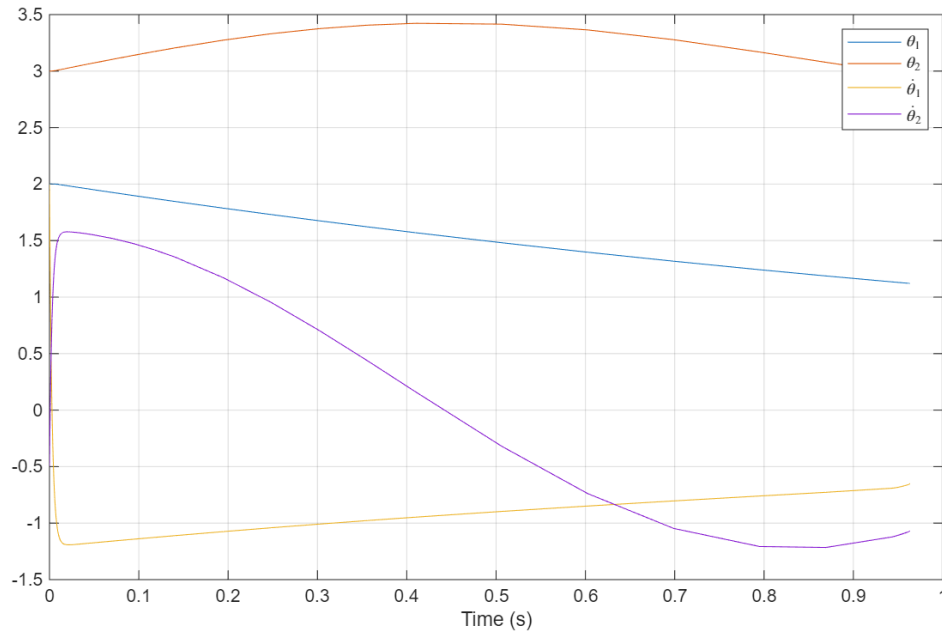


Figure 14: Closed-loop system behavior with incompatible ISS-ISSf-QP controller. No disturbance simulated.

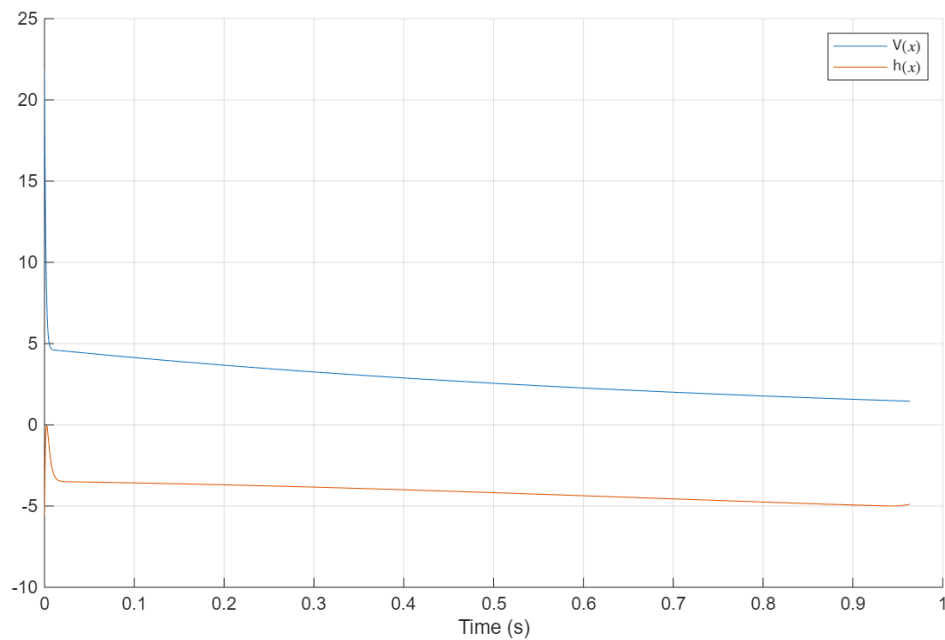


Figure 15: $V(x)$ and $h(x)$ with incompatible ISS-ISSf-QP controller.

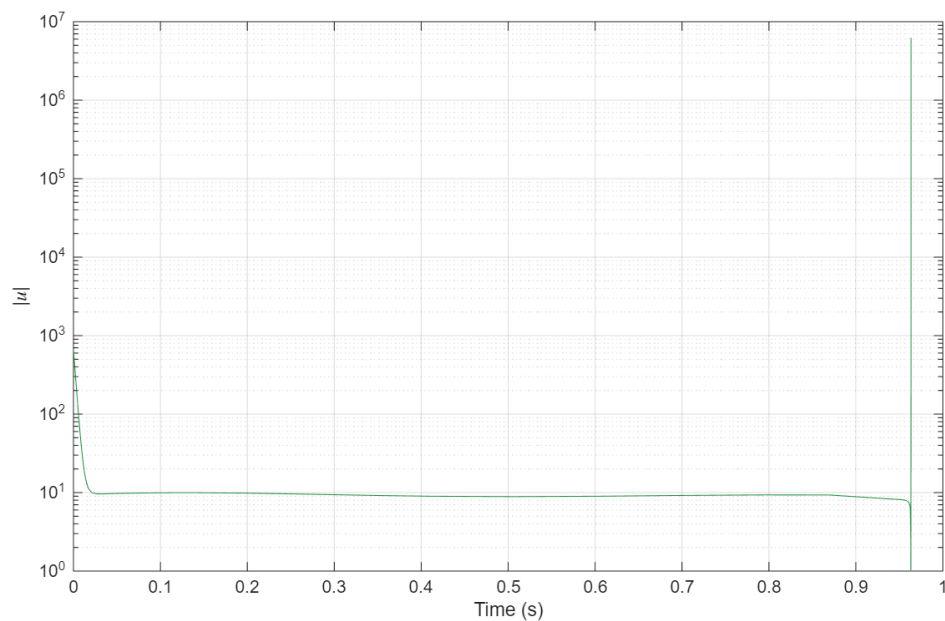


Figure 16: Magnitude of control input of incompatible ISS-ISSf-QP controller.

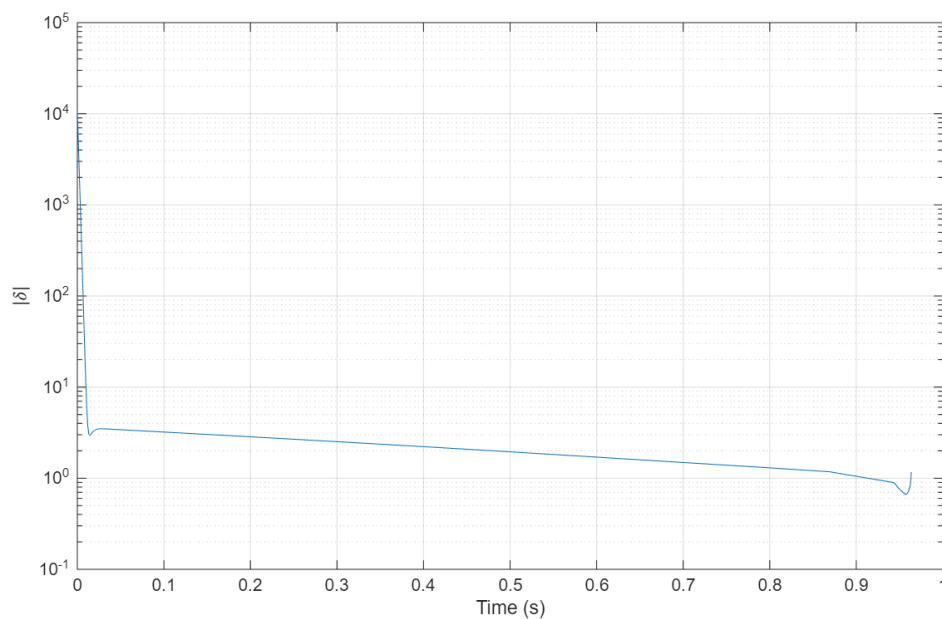


Figure 17: Magnitude of δ of incompatible ISS-ISSf-QP controller.

It appears from Figure ?? that the controller is primarily trying to achieve asymptotic stability, as $V(x)$ is driven towards zero. However, if we further examine the plot, we see that at no point does the system enter the safe set $\mathcal{S}'$, i.e. $h(x)$ never exceeds 0. This is also supported by Figure 17, where the value of δ does not get near zero. The incompatible conditions are likely what drives the control input in Figure 16 towards infinity, which causes the failure of the quadratic program solver.

4 Appendix

4.1 System

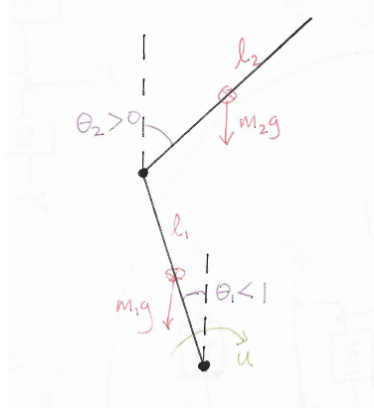


Figure 18: Double inverted pendulum.

An input torque u is applied to the first link of the pendulum. Each arm is modeled as a uniform rod ($I_i = \frac{1}{12}m_i l_i^2$) and friction between the rods is assumed as $|F_1| = b_1|\dot{\theta}_1|$ and $|F_2| = b_2|\dot{\theta}_1 - \dot{\theta}_2|$. Using the angular deviation from the vertical and angular velocities of the links, we can define the state $x = [\theta_1 \ \theta_2 \ \dot{\theta}_1 \ \dot{\theta}_2]^\top$. The dynamics of the system are driven by the following differential equation in the form $\dot{x} = f(x, u)$:

$$f(x, u) = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & \mu_2(x) & \mu_3(x) \\ 0 & 0 & \mu_3(x) & \mu_1(x) \end{bmatrix} \left(\begin{bmatrix} \nu_1(x) \\ \nu_2(x) \\ \nu_3(x) \\ \nu_4(x) \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ 1 \\ 0 \end{bmatrix} u \right) = \begin{bmatrix} \nu_1(x) \\ \nu_2(x) \\ \mu_2(x)\nu_3(x) + \mu_3(x)\nu_4(x) \\ \mu_3(x)\nu_3(x) + \mu_1(x)\nu_4(x) \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ \mu_2(x) \\ \mu_3(x) \end{bmatrix} u$$

4.2 System equations

$$\begin{aligned} \mu_2(x) &= \frac{M_2}{|M(x)|}, \quad \mu_3(x) = \frac{-M_3 \cos \Delta}{|M(x)|}, \quad \mu_1(x) = \frac{M_1}{|M(x)|} \\ \nu_1(x) &= \dot{\theta}_1, \quad \nu_2(x) = \dot{\theta}_2 \\ \nu_3(x) &= -K_1 \dot{\theta}_2^2 \sin \Delta + K_2 \sin \theta_1 - B_1 \dot{\theta}_1 + B_2 \dot{\theta}_2 \\ \nu_4(x) &= K_3 \dot{\theta}_1^2 \sin \Delta + K_4 \sin \theta_2 + B_2 \dot{\theta}_1 - B_2 \dot{\theta}_2 \end{aligned}$$

with parameters defined in Table 1.

M_1	$(\frac{1}{3}m_1 + m_2) l_1^2$
M_2	$\frac{1}{3}m_2 l_2^2$
M_3	$\frac{1}{2}m_2 l_1 l_2$
K_1	$\frac{1}{2}m_2 l_1 l_2$
K_2	$(\frac{1}{2}m_1 + m_2) g l_1$
K_3	$\frac{1}{2}m_2 l_1 l_2$
K_4	$\frac{1}{2}m_2 g l_2$
B_1	$(b_1 + b_2)$
B_2	b_2
Δ	$\theta_1 - \theta_2$
$ M(x) $	$M_1 M_2 - M_3^2 \cos^2 \Delta$

Table 1: Coefficients for rest of derivations.

4.3 Jacobian

The Jacobian of the system is:

$$Df(x) = \begin{bmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ Df_{31}(x) & Df_{32}(x) & Df_{33}(x) & Df_{34}(x) \\ Df_{41}(x) & Df_{42}(x) & Df_{43}(x) & Df_{44}(x) \end{bmatrix}$$

where the elements are defined by the equations below:

$$\begin{aligned} Df_{31}(x) &= \frac{\partial \mu_2(x)}{\partial \theta_1} \nu_3(x) + \mu_2(x) \frac{\partial \nu_3(x)}{\partial \theta_1} + \frac{\partial \mu_3(x)}{\partial \theta_1} \nu_4(x) + \mu_3(x) \frac{\partial \nu_4(x)}{\partial \theta_1} \\ Df_{32}(x) &= \frac{\partial \mu_2(x)}{\partial \theta_2} \nu_3(x) + \mu_2(x) \frac{\partial \nu_3(x)}{\partial \theta_2} + \frac{\partial \mu_3(x)}{\partial \theta_2} \nu_4(x) + \mu_3(x) \frac{\partial \nu_4(x)}{\partial \theta_2} \\ Df_{33}(x) &= \mu_2(x) \frac{\partial \nu_3(x)}{\partial \dot{\theta}_1} + \mu_3(x) \frac{\partial \nu_4(x)}{\partial \dot{\theta}_1} \\ Df_{34}(x) &= \mu_2(x) \frac{\partial \nu_3(x)}{\partial \dot{\theta}_2} + \mu_3(x) \frac{\partial \nu_4(x)}{\partial \dot{\theta}_2} \\ Df_{41}(x) &= \frac{\partial \mu_3(x)}{\partial \theta_1} \nu_3(x) + \mu_3(x) \frac{\partial \nu_3(x)}{\partial \theta_1} + \frac{\partial \mu_1(x)}{\partial \theta_1} \nu_4(x) + \mu_1(x) \frac{\partial \nu_4(x)}{\partial \theta_1} \\ Df_{42}(x) &= \frac{\partial \mu_3(x)}{\partial \theta_2} \nu_3(x) + \mu_3(x) \frac{\partial \nu_3(x)}{\partial \theta_2} + \frac{\partial \mu_1(x)}{\partial \theta_2} \nu_4(x) + \mu_1(x) \frac{\partial \nu_4(x)}{\partial \theta_2} \\ Df_{43}(x) &= \mu_3(x) \frac{\partial \nu_3(x)}{\partial \dot{\theta}_1} + \mu_1(x) \frac{\partial \nu_4(x)}{\partial \dot{\theta}_1} \\ Df_{44}(x) &= \mu_3(x) \frac{\partial \nu_3(x)}{\partial \dot{\theta}_2} + \mu_1(x) \frac{\partial \nu_4(x)}{\partial \dot{\theta}_2} \end{aligned}$$

$$D \begin{bmatrix} \mu_1(x) \\ \mu_2(x) \\ \mu_3(x) \end{bmatrix} = \begin{bmatrix} -M_1 M_3 \frac{\cos \Delta \sin \Delta}{|M(x)|^2} & M_1 M_3 \frac{\cos \Delta \sin \Delta}{|M(x)|^2} & 0 & 0 \\ -M_2 M_3 \frac{\cos \Delta \sin \Delta}{|M(x)|^2} & M_2 M_3 \frac{\cos \Delta \sin \Delta}{|M(x)|^2} & 0 & 0 \\ -M_3 \frac{(2M_3 \cos^2 \Delta - |M(x)|) \sin \Delta}{|M(x)|^2} & M_3 \frac{(2M_3 \cos^2 \Delta - |M(x)|) \sin \Delta}{|M(x)|^2} & 0 & 0 \end{bmatrix}$$

$$D \begin{bmatrix} \nu_1(x) \\ \nu_2(x) \\ \nu_3(x) \\ \nu_4(x) \end{bmatrix} = \begin{bmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ -K_1 \dot{\theta}_2^2 \cos \Delta + K_2 \cos \theta_1 & K_1 \dot{\theta}_2^2 \cos \Delta & -B_1 & -2K_1 \dot{\theta}_2 \sin \Delta + B_2 \\ K_3 \dot{\theta}_1^2 \cos \Delta & -K_3 \dot{\theta}_1^2 \cos \Delta + K_4 \cos \theta_2 & 2K_3 \dot{\theta}_1 \sin \Delta + B_2 & -B_2 \end{bmatrix}$$

4.4 Meme

